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Henning et al.

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(54) **DUAL PURPOSE ENERGY PLANT**

(71) Applicant: **NUOVO PIGNONE TECNOLOGIE S.r.l.**, Florence (IT)

(72) Inventors: **Carlos Henning**, Buenos Aires (AR); **David Madden**, Limerick (IE)

(73) Assignee: **NUOVO PIGNONE TECNOLOGIE S.R.L.**, Florence (IT)

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F01N 3/08 (2006.01)
F01N 5/02 (2006.01)

(52) **U.S. Cl.**

CPC ***F01D 15/10*** (2013.01); ***F01N 3/0857*** (2013.01); ***F01N 3/0871*** (2013.01); ***F01N 5/02*** (2013.01); *F01N 2270/10* (2013.01); *F01N 2310/00* (2013.01); *F05D 2220/31* (2013.01); *F05D 2260/61* (2013.01)

(58) **Field of Classification Search**

CPC F01D 15/10; F01N 3/0857; F01N 3/0871; F01N 5/02; F01N 2270/10; F01N 2310/00; F05D 2220/31; F05D 2260/61
See application file for complete search history.

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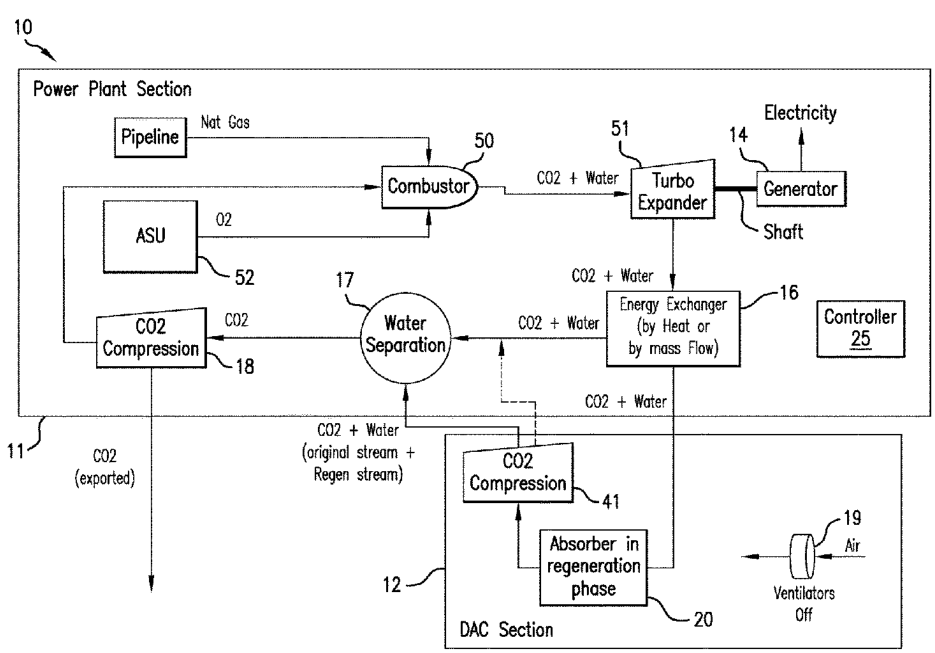
Primary Examiner — Jesse S Bogue
(74) *Attorney, Agent, or Firm* — CANTOR COLBURN LLP

(57) **ABSTRACT**

A system for generating electricity with reduced or negative carbon emissions. The system includes a power plant section having an electricity generating unit having an input coupled to a hydrocarbon fuel supply and an energy exchange path. The system also includes a direct air capture (DAC) section having a CO_2 adsorption device having a CO_2 adsorbent material and a ventilator electrically coupled to the electricity generating unit, the ventilator directing air flow through the CO_2 adsorption device in a carbon capture mode, wherein the CO_2 adsorption device is coupled to and in energy communication with the energy exchange path for releasing adsorbed CO_2 in a carbon release mode.

9 Claims, 15 Drawing Sheets



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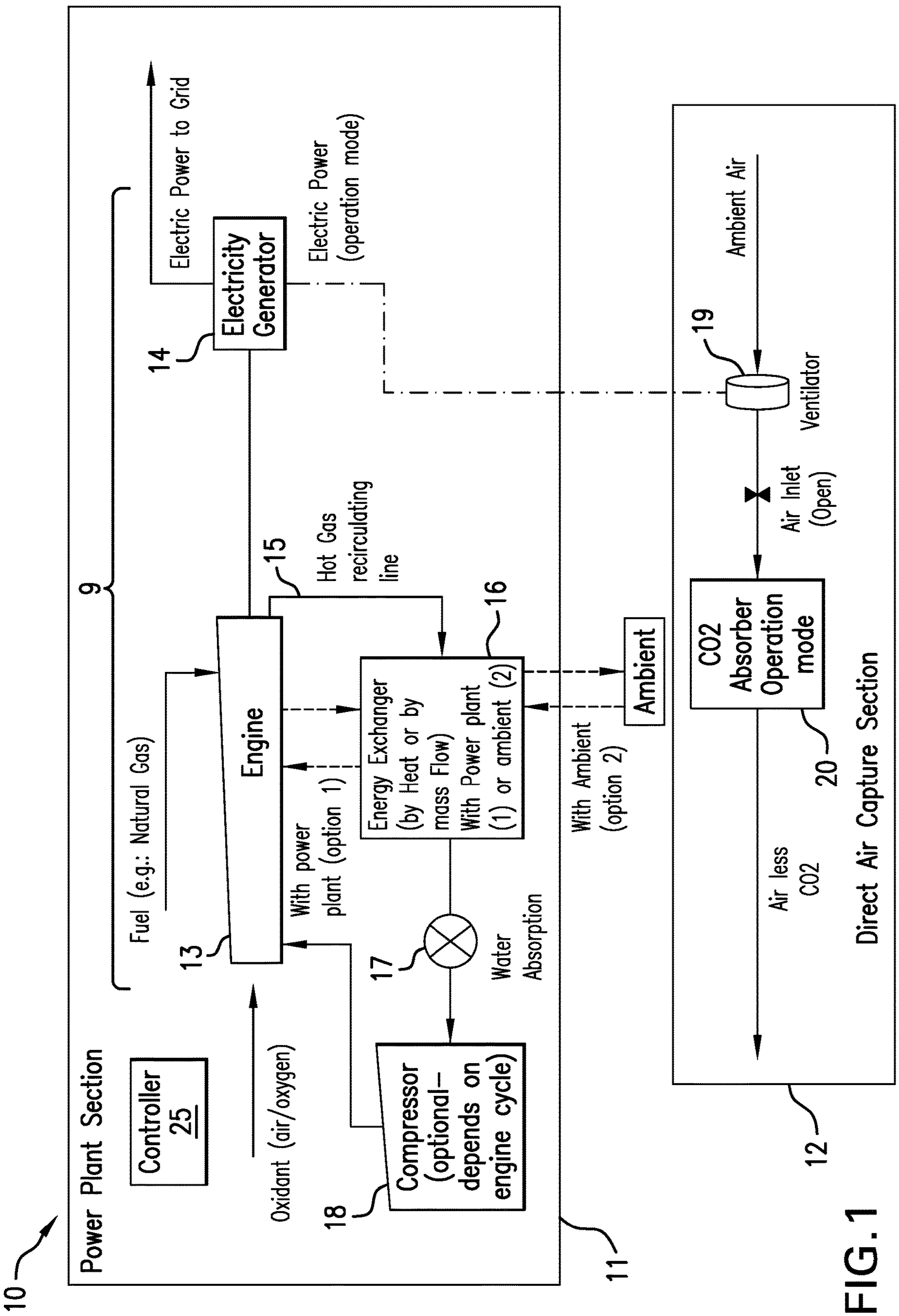

10
Power Plant Section
Controller
25
Oxidant (air/oxygen)
Compressor (optional— depends on engine cycle)
18
11
13
Engine
Fuel (e.g.: Natural Gas)
9
With power plant (option 1)
17
Water Absorption
Energy Exchanger (by Heat or by mass Flow) With Power plant (1) or ambient (2)
16
15
Hot Gas recirculating line
With Ambient (option 2)
Ambient
14
Electricity Generator
Electric Power to Grid
Electric Power (operation mode)
Direct Air Capture Section
Ambient Air
19
Ventilator
Air Inlet (Open)
CO2 Absorber Operation mode
20
Air less CO2
12

FIG. 1

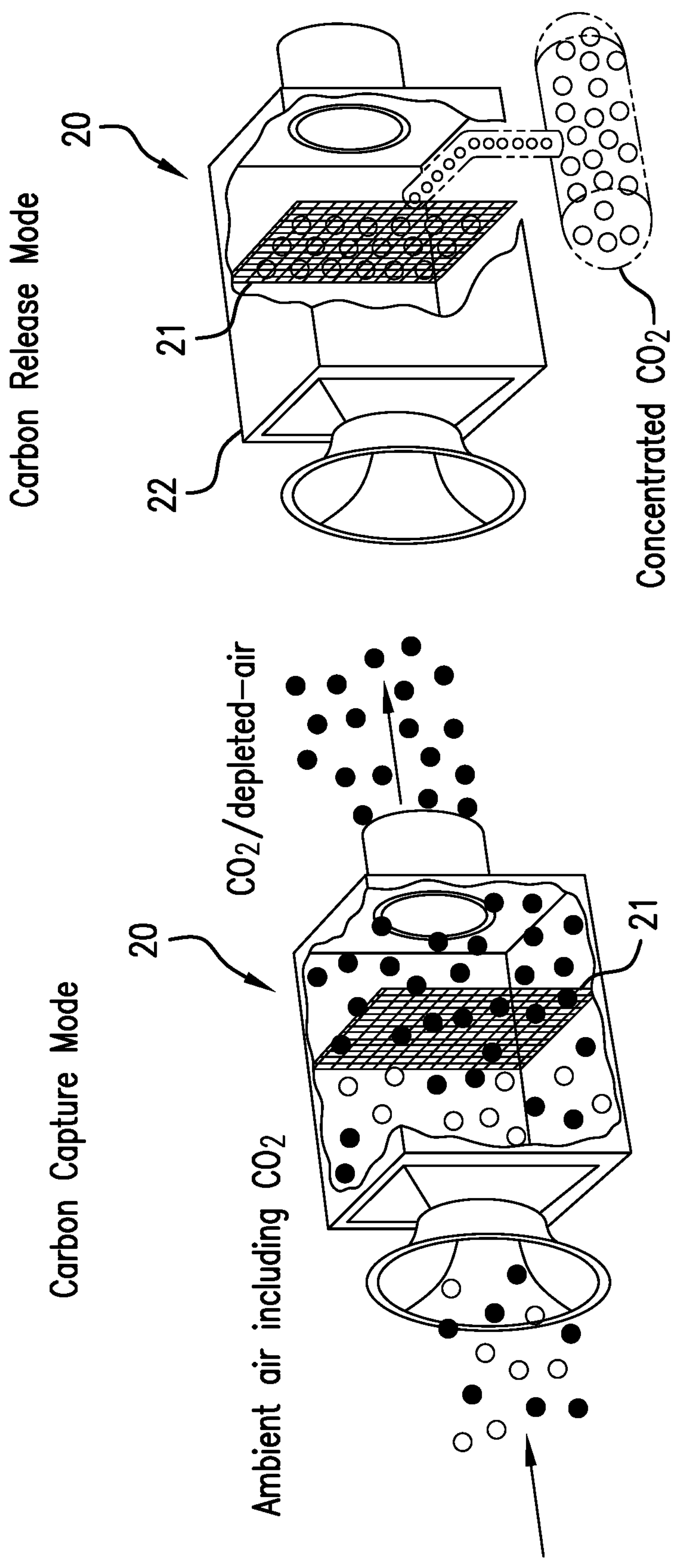

Carbon Release Mode
20
21
22
Concentrated CO_2
Carbon Capture Mode
20
21
CO_2/depleted-air
Ambient air including CO_2

FIG.2

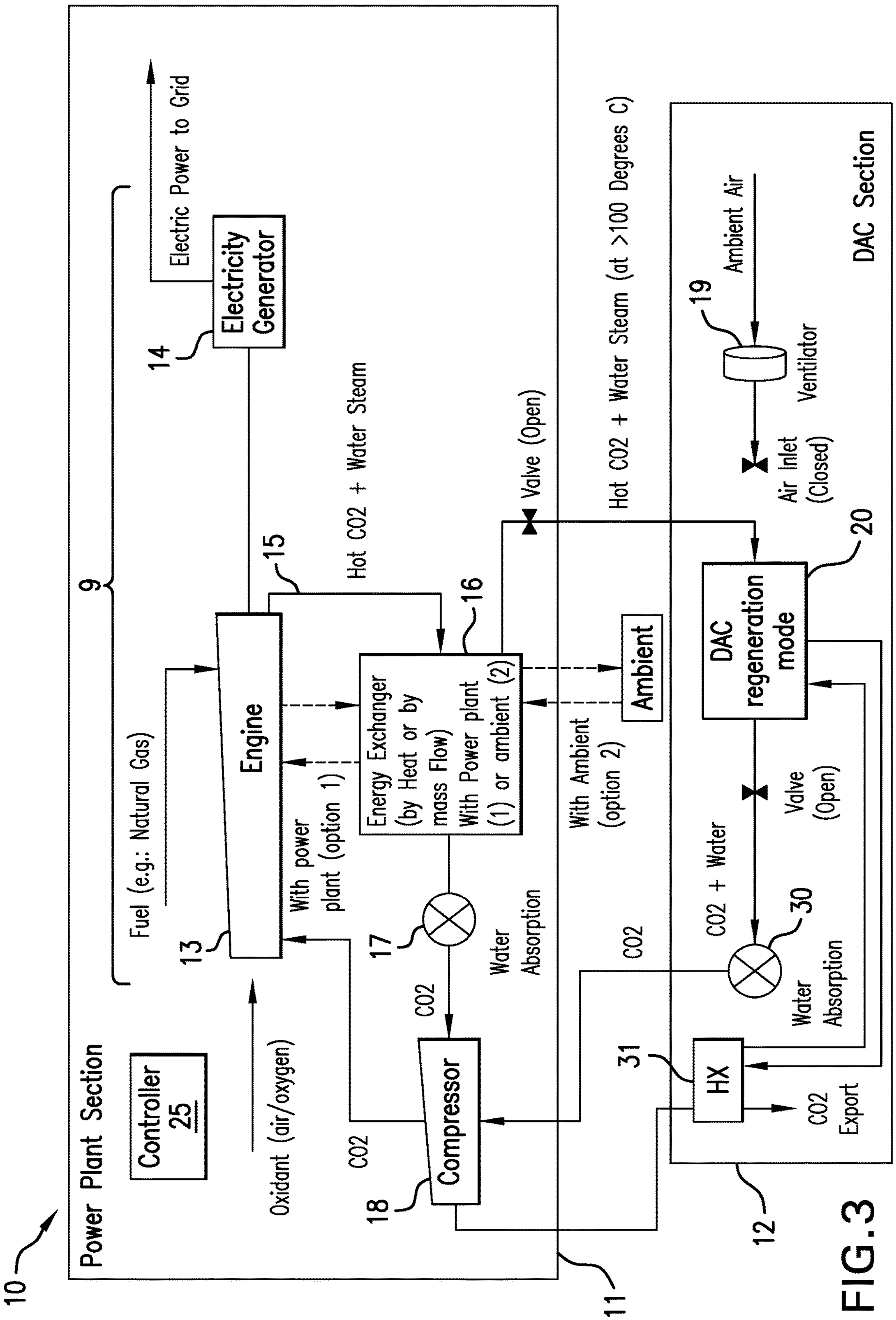
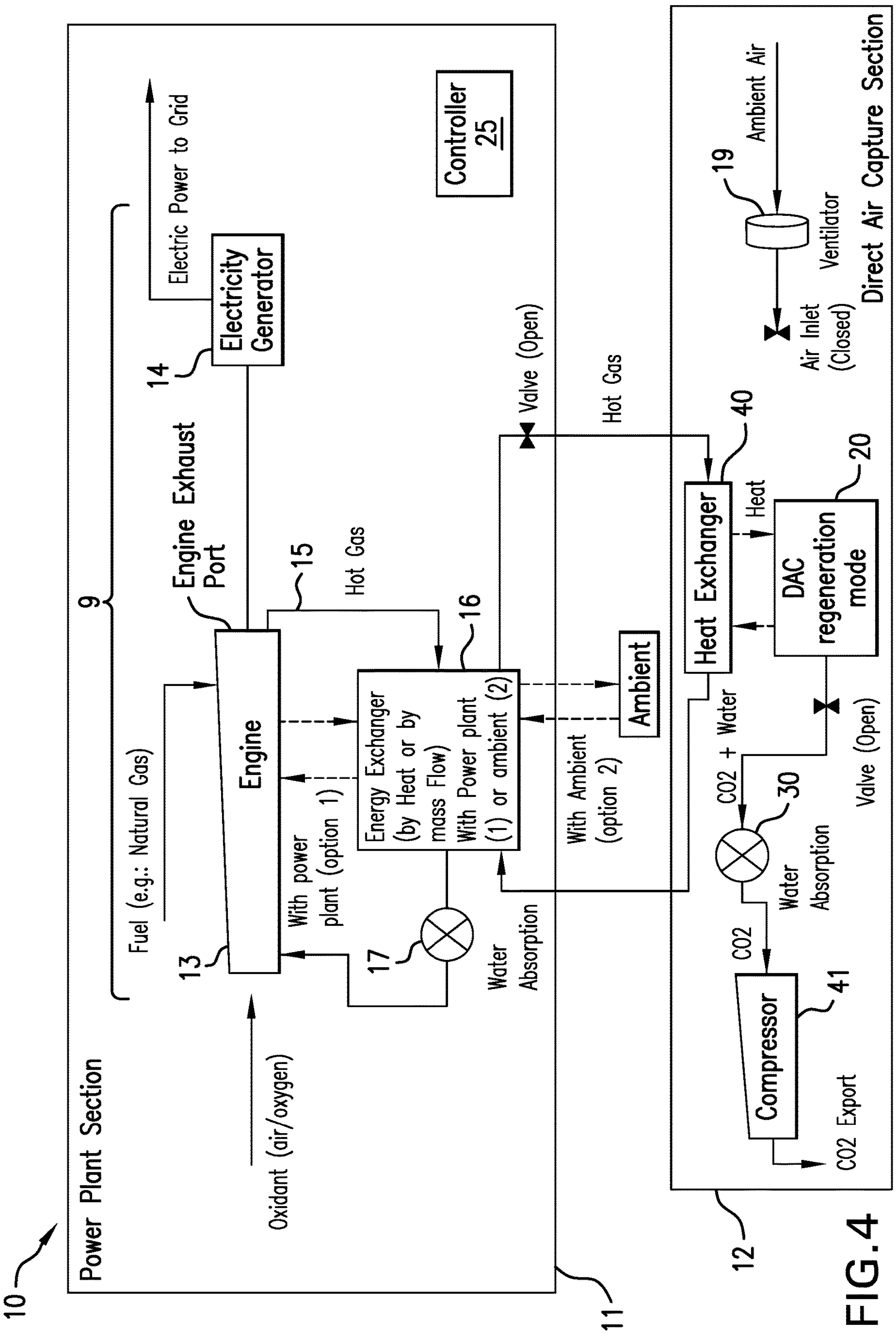
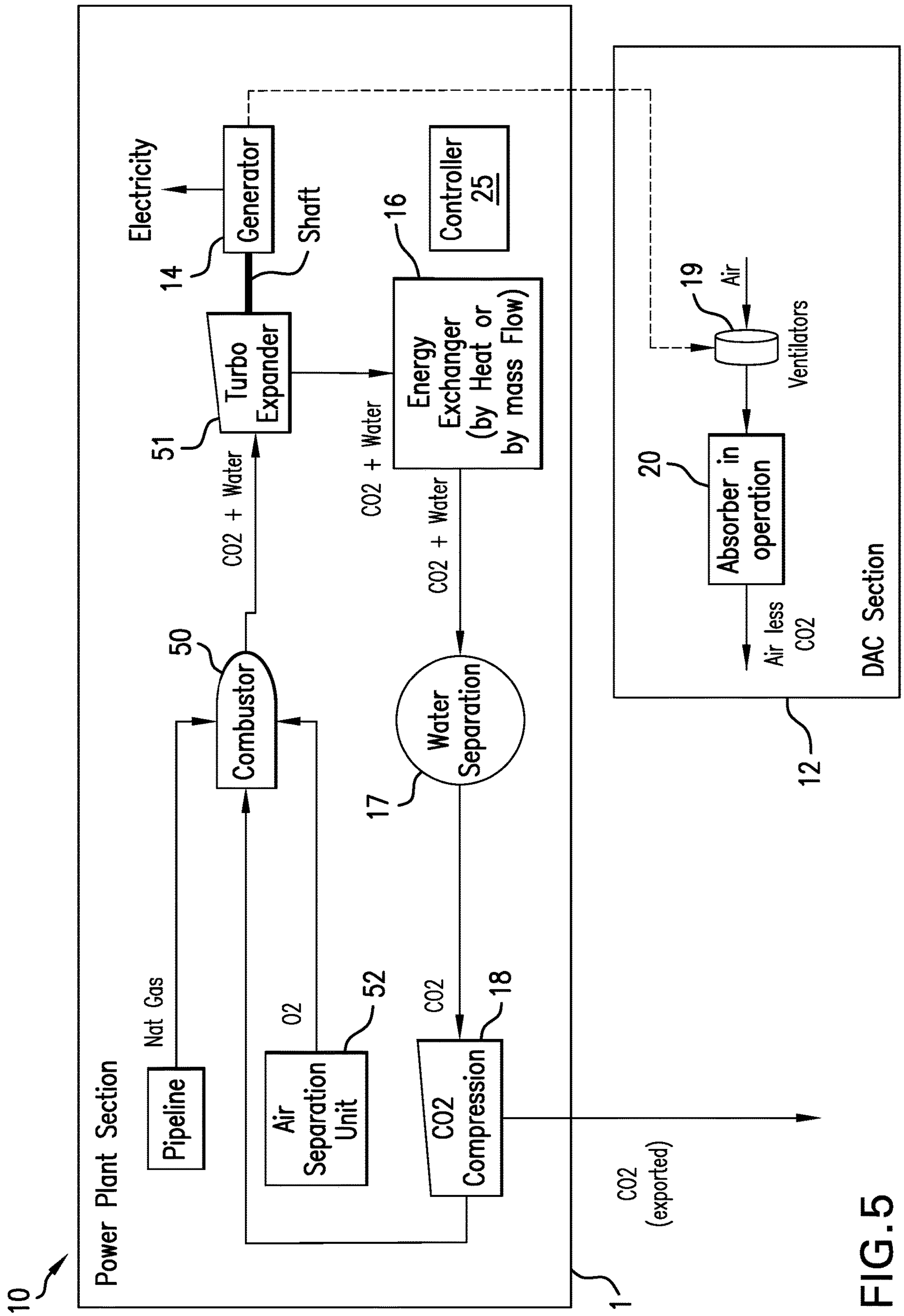

10
Power Plant Section
Controller
25
Oxidant (air/oxygen)
CO2
18
Compressor
13
Fuel (e.g.: Natural Gas)
Engine
9
With power
plant (option 1)
17
CO2
Water
Absorption
15
Hot CO2 + Water Steam
16
Energy Exchanger
(by Heat or by
mass Flow)
With Power plant
(1) or ambient (2)
Ambient
With Ambient
(option 2)
14
Electricity
Generator
Electric Power to Grid
Valve (Open)
Hot CO2 + Water Steam (at >100 Degrees C)
11
12
DAC Section
Ambient Air
19
Ventilator
Air Inlet
(Closed)
20
DAC
regeneration
mode
Valve
(Open)
CO2 + Water
30
Water
Absorption
CO2
31
HX
CO2
Export

FIG.3


10
Power Plant Section
11
9
Fuel (e.g.: Natural Gas)
13
Engine
Engine Exhaust Port
Oxidant (air/oxygen)
With power plant (option 1)
14
Electricity Generator
Electric Power to Grid
Controller 25
15
Hot Gas
16
Energy Exchanger (by Heat or by mass Flow) With Power plant (1) or ambient (2)
17
Water Absorption
Ambient
With Ambient (option 2)
Valve (Open)
Hot Gas
12
Direct Air Capture Section
40
Heat Exchanger
Heat
20
DAC regeneration mode
19
Ventilator
Ambient Air
Air Inlet (Closed)
CO2 + Water
Valve (Open)
30
Water Absorption
CO2
41
Compressor
CO2 Export
FIG. 4



10
Power Plant Section
11
Pipeline
Nat Gas
Air Separation Unit
52
O2
CO2 Compression
18
CO2
CO2 (exported)
Combustor
50
CO2 + Water
Turbo Expander
51
Shaft
Generator
14
Electricity
Energy Exchanger (by Heat or by mass Flow)
16
CO2 + Water
CO2 + Water
Water Separation
17
Controller 25
Air
19
Ventilators
Absorber in operation
20
Air less CO2
DAC Section
12

FIG. 5

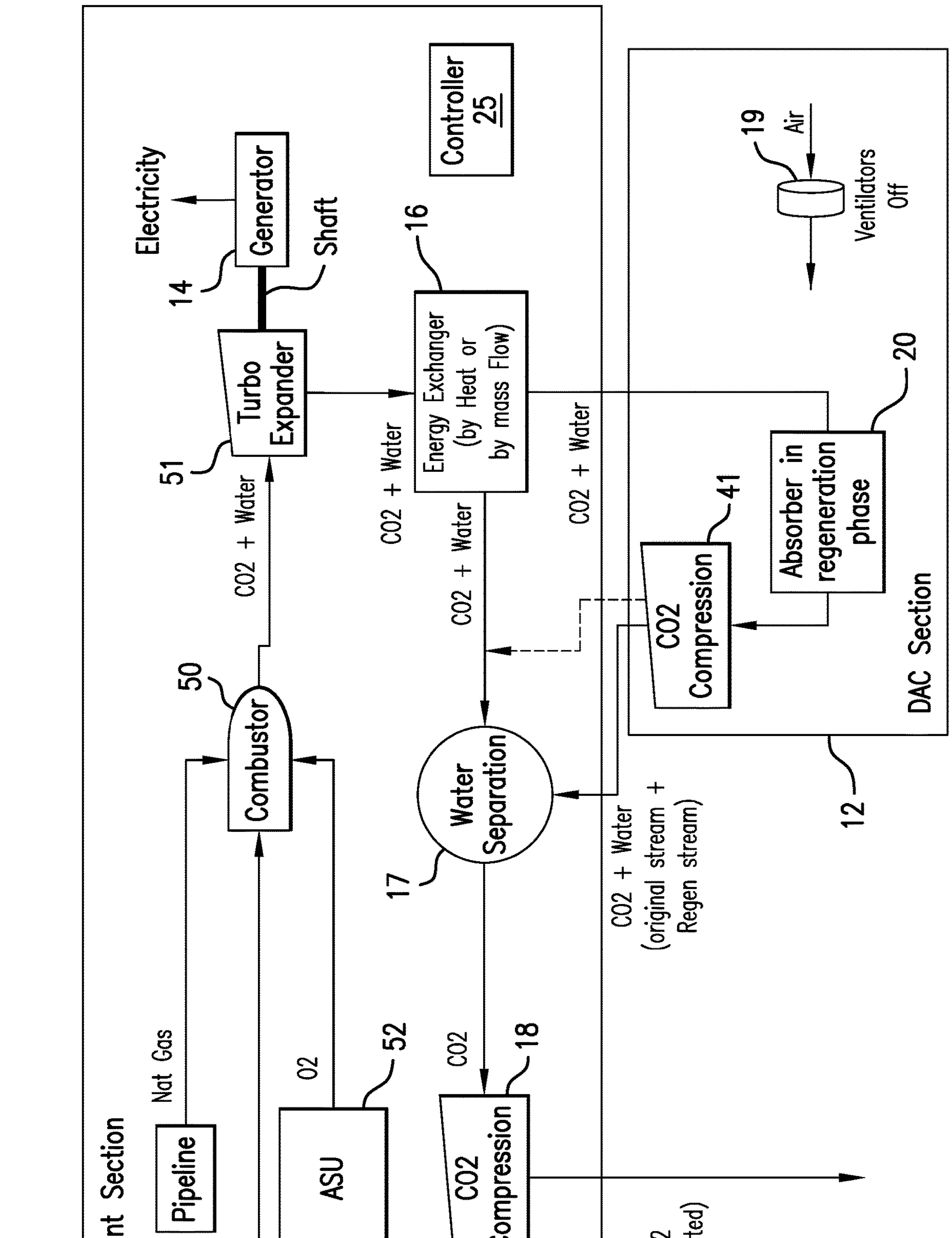

10
Power Plant Section
11
Pipeline
Nat Gas
ASU
O2
52
CO2 Compression
CO2
18
CO2 (exported)
Combustor
50
CO2 + Water
Turbo Expander
51
Shaft
Generator
14
Electricity
Energy Exchanger (by Heat or by mass Flow)
16
CO2 + Water
CO2 + Water
CO2 + Water
Water Separation
17
CO2 + Water (original stream + Regen stream)
Controller 25
DAC Section
12
CO2 Compression
41
Absorber in regeneration phase
20
19
Air
Ventilators Off

FIG.6

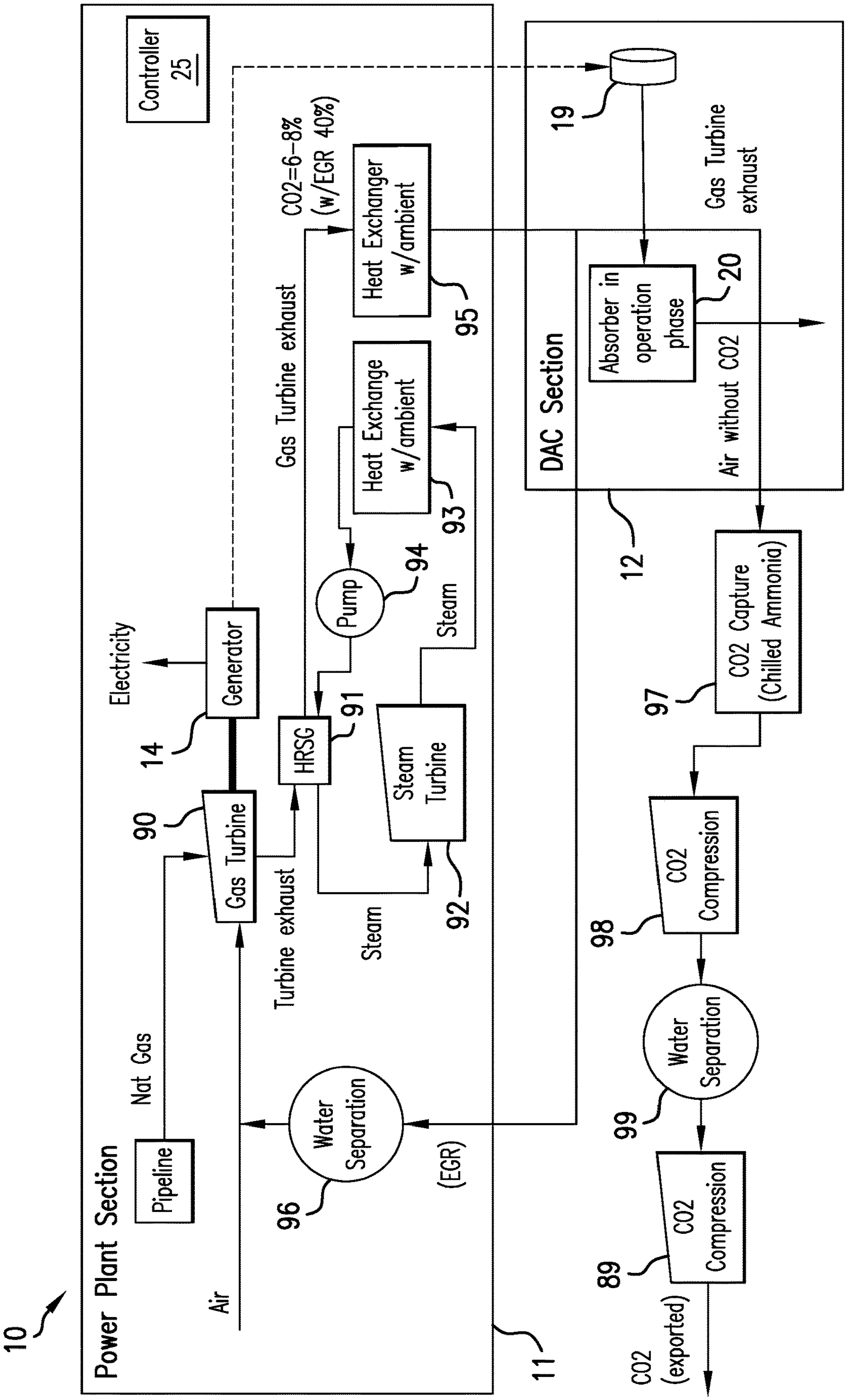
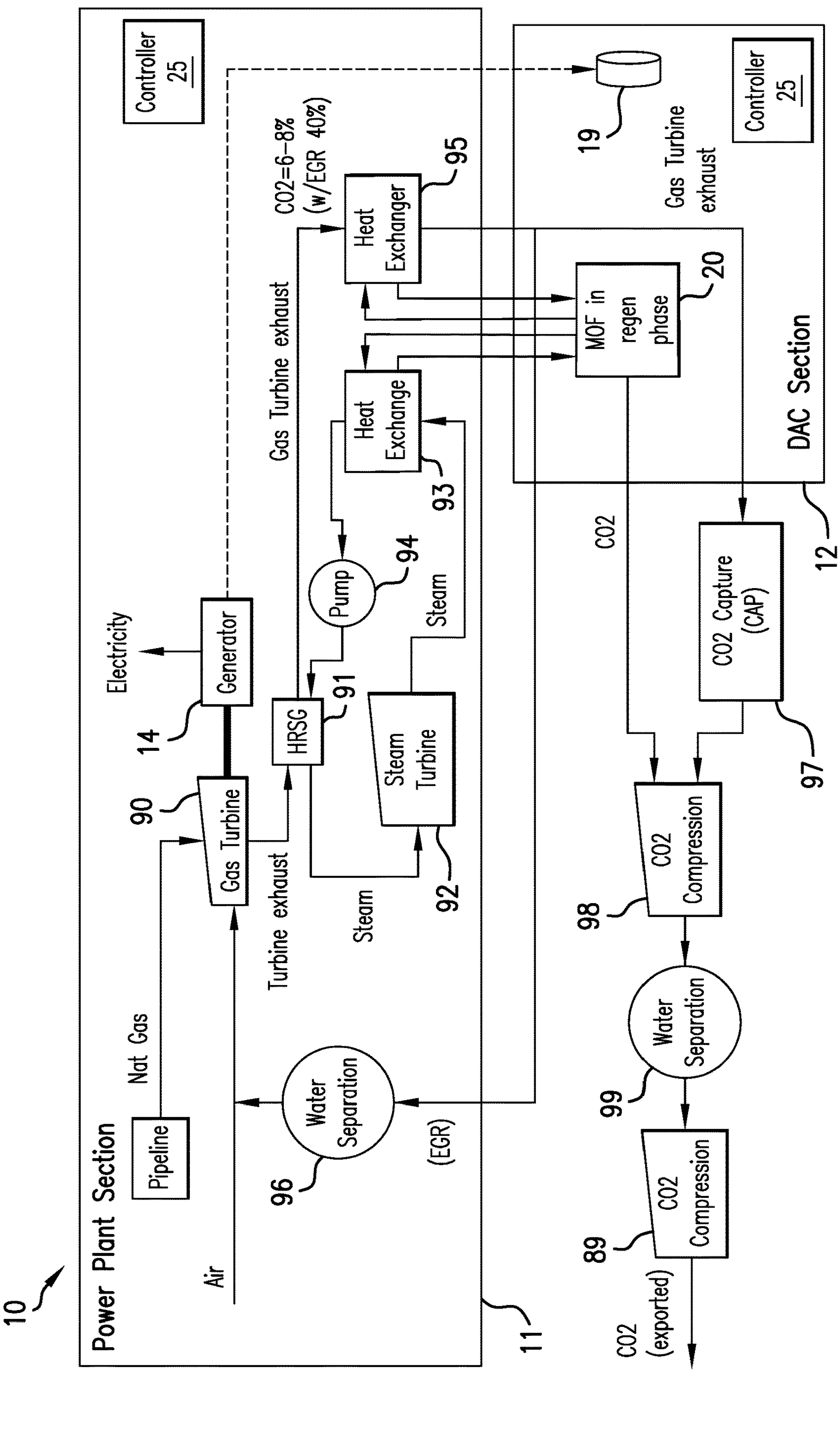

10
Power Plant Section
11
Controller
25
Pipeline
Nat Gas
Air
Gas Turbine
90
Generator
14
Electricity
HRSG
91
Steam Turbine
92
Steam
Turbine exhaust
Pump
94
Heat Exchange w/ambient
93
Steam
Gas Turbine exhaust
Heat Exchanger w/ambient
95
CO2=6–8% (w/EGR 40%)
Water Separation
96
(EGR)
DAC Section
12
19
Absorber in operation phase
20
Gas Turbine exhaust
Air without CO2
CO2 Capture (Chilled Ammonia)
97
CO2 Compression
98
Water Separation
99
CO2 Compression
89
CO2 (exported)

FIG. 7



10
Power Plant Section
11
Controller
25
Pipeline
Nat Gas
Air
Gas Turbine
90
Generator
14
Electricity
Turbine exhaust
HRSG
91
Steam
Steam Turbine
92
Steam
Pump
94
Heat Exchange
93
Gas Turbine exhaust
Heat Exchanger
95
CO2=6-8%
(w/EGR 40%)
Water Separation
96
(EGR)
DAC Section
12
Controller
25
19
Gas Turbine exhaust
MOF in regen phase
20
CO2
CO2 Capture (CAP)
97
CO2 Compression
98
Water Separation
99
CO2 Compression
89
CO2 (exported)

FIG. 8

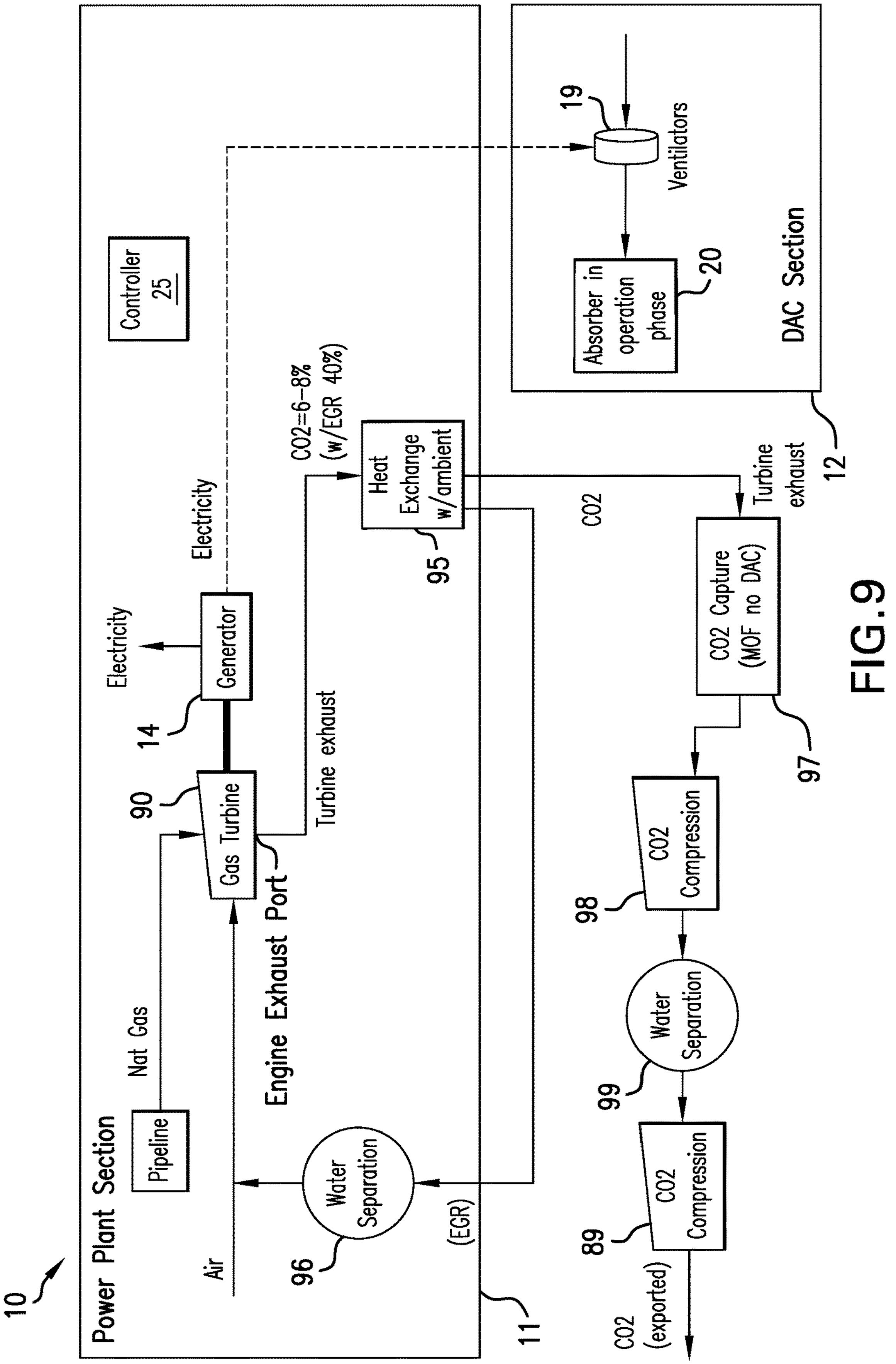

10
Power Plant Section
11
Pipeline
Nat Gas
Air
Controller
25
Gas Turbine
90
Generator
14
Electricity
Electricity
Engine Exhaust Port
Turbine exhaust
CO2=6-8%
(w/EGR 40%)
Heat
Exchange
w/ambient
95
CO2
Water
Separation
96
(EGR)
DAC Section
12
19
Ventilators
Absorber in
operation
phase
20
Turbine
exhaust
CO2 Capture
(MOF no DAC)
97
CO2
Compression
98
Water
Separation
99
CO2
Compression
89
CO2
(exported)

FIG.9

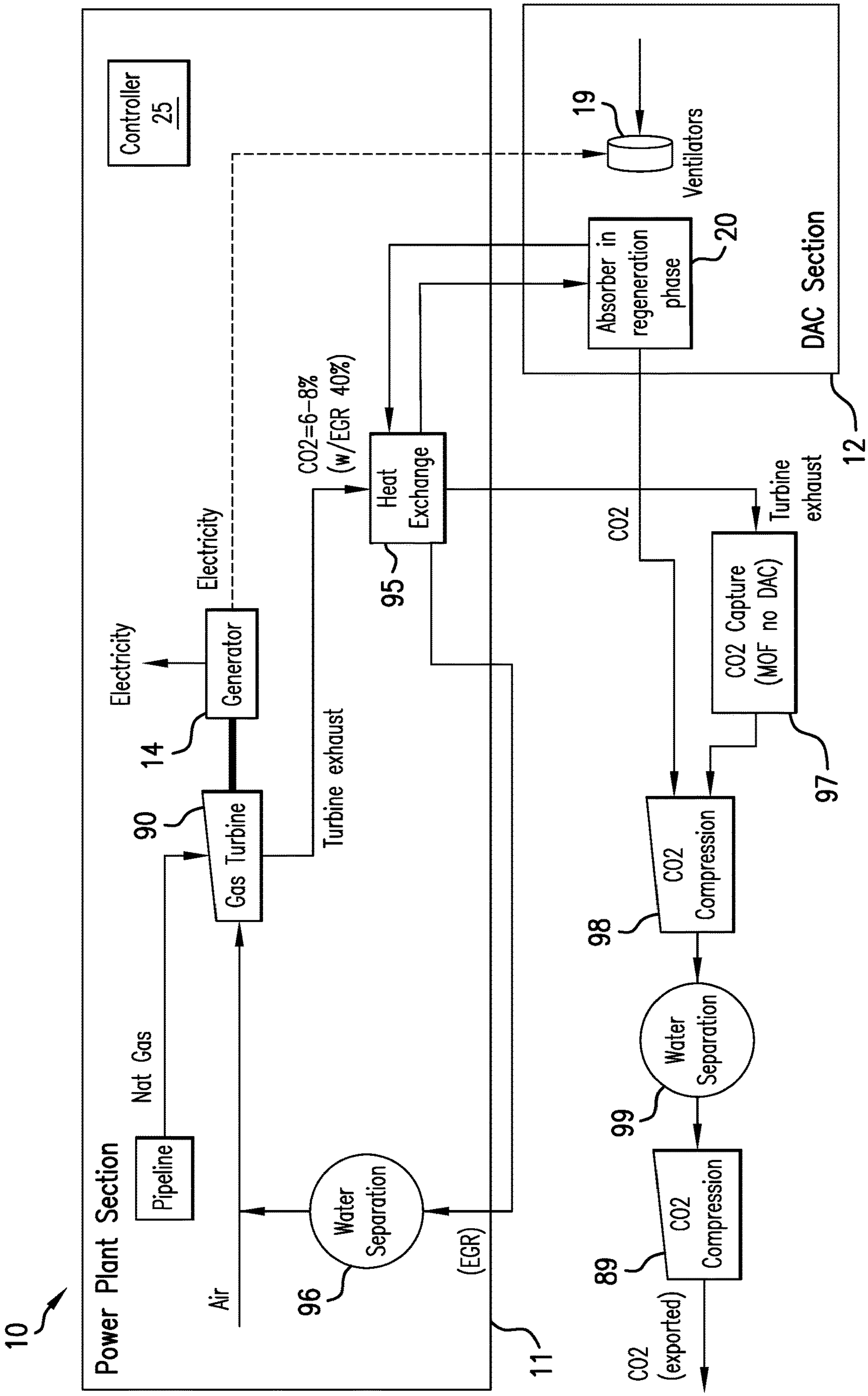

10
Power Plant Section
11
Controller
25
Pipeline
Nat Gas
Air
Gas Turbine
90
Generator
14
Electricity
Electricity
Turbine exhaust
Water Separation
96
(EGR)
CO2=6-8%
(w/EGR 40%)
Heat Exchange
95
DAC Section
12
19
Ventilators
Absorber in regeneration phase
20
CO2
Turbine exhaust
CO2 Capture (MOF no DAC)
97
CO2 Compression
98
Water Separation
99
CO2 Compression
89
CO2 (exported)

FIG.10

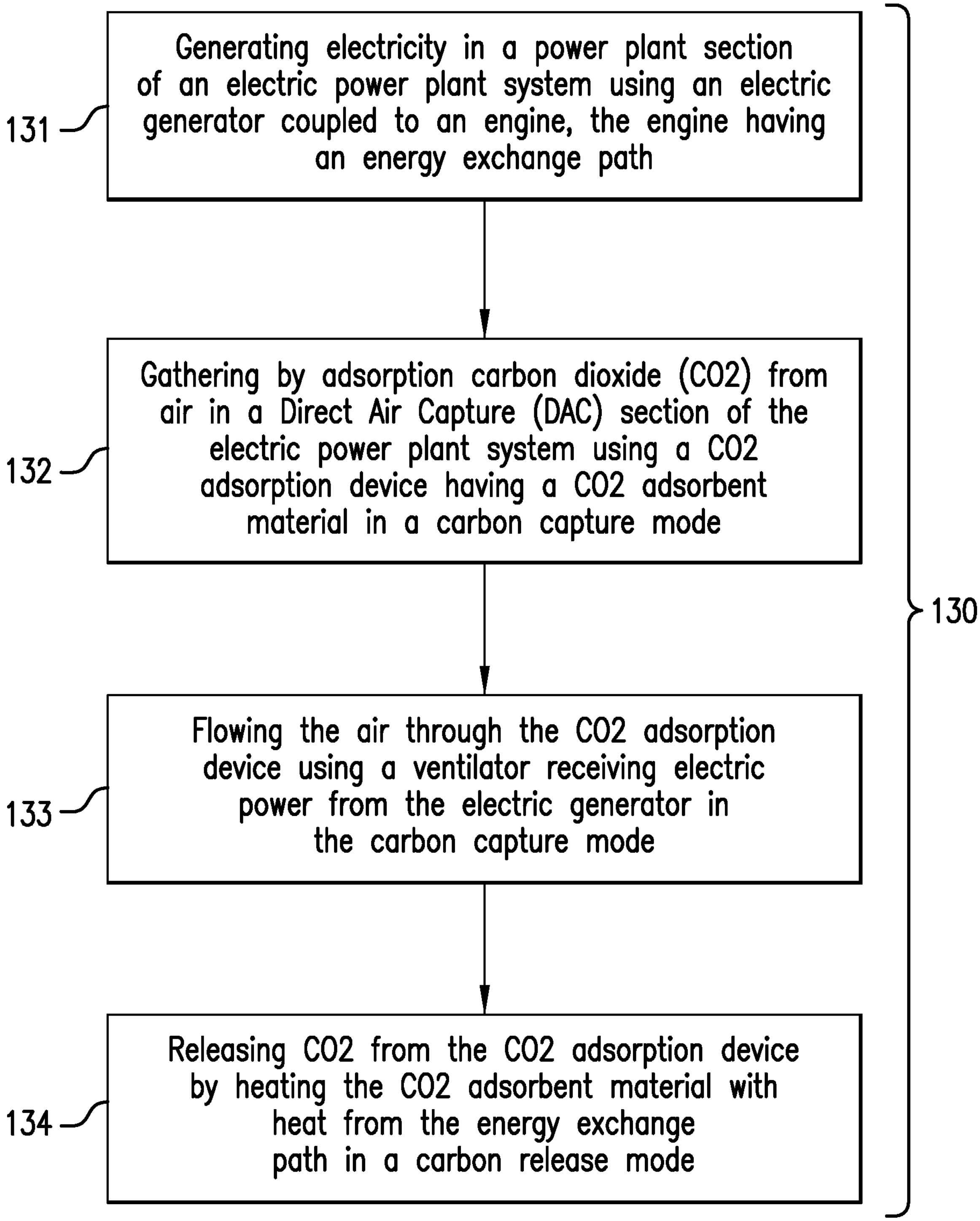
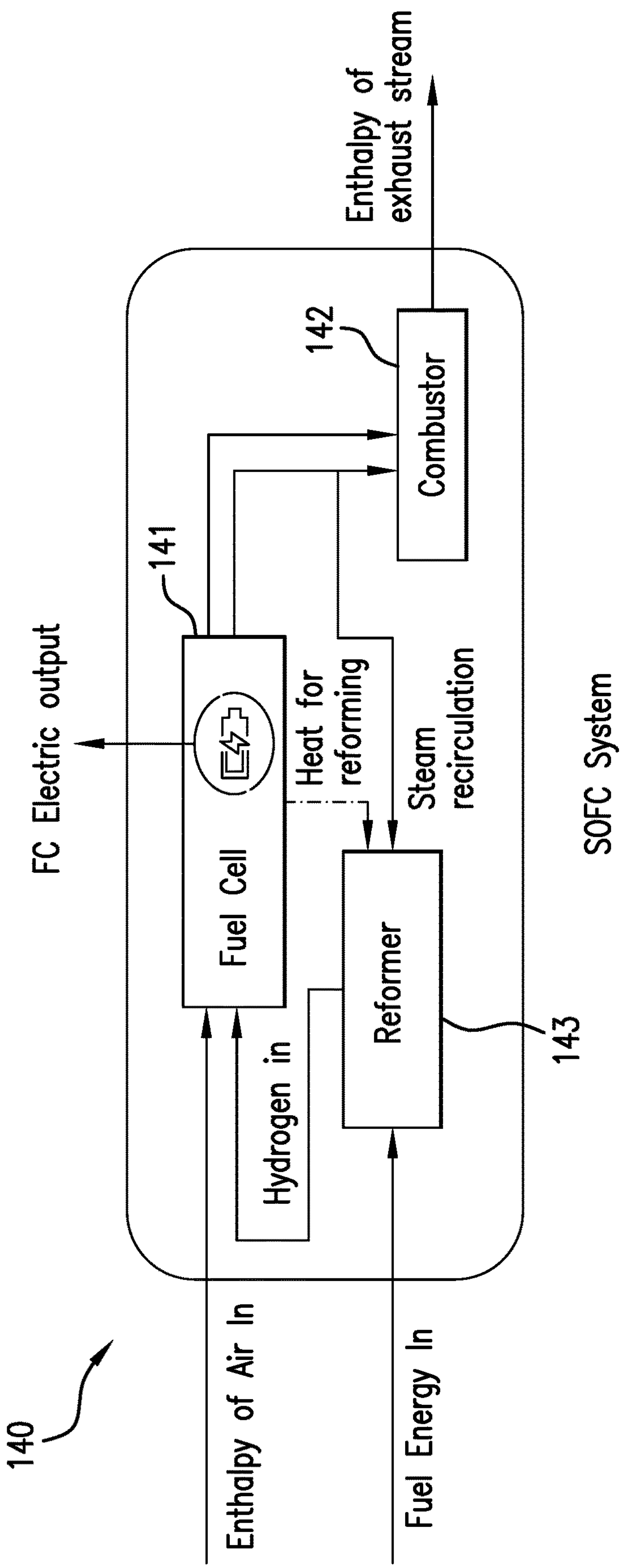

131
Generating electricity in a power plant section of an electric power plant system using an electric generator coupled to an engine, the engine having an energy exchange path
132
Gathering by adsorption carbon dioxide (CO2) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO2 adsorption device having a CO2 adsorbent material in a carbon capture mode
133
Flowing the air through the CO2 adsorption device using a ventilator receiving electric power from the electric generator in the carbon capture mode
134
Releasing CO2 from the CO2 adsorption device by heating the CO2 adsorbent material with heat from the energy exchange path in a carbon release mode
130

FIG. 11



$CH_{4,\,anode} + 2H_2O_{,\,anode} \longrightarrow 4H_{2,\,anode}\ CO_{2,\,anode}$ (Steam Reforming Reaction)

$4H_{2,\,anode} + 2O_{2,\,cathode} \longrightarrow 4H_2O_{,\,anode}$ (Fuel Cell Reaction)

$CH_{4,\,anode} + 2O_{2,\,cathode} \longrightarrow 2H_2O_{,\,anode}\ CO_{2,\,anode}$ (Combined Reforming and FC Reactions)

FIG. 12

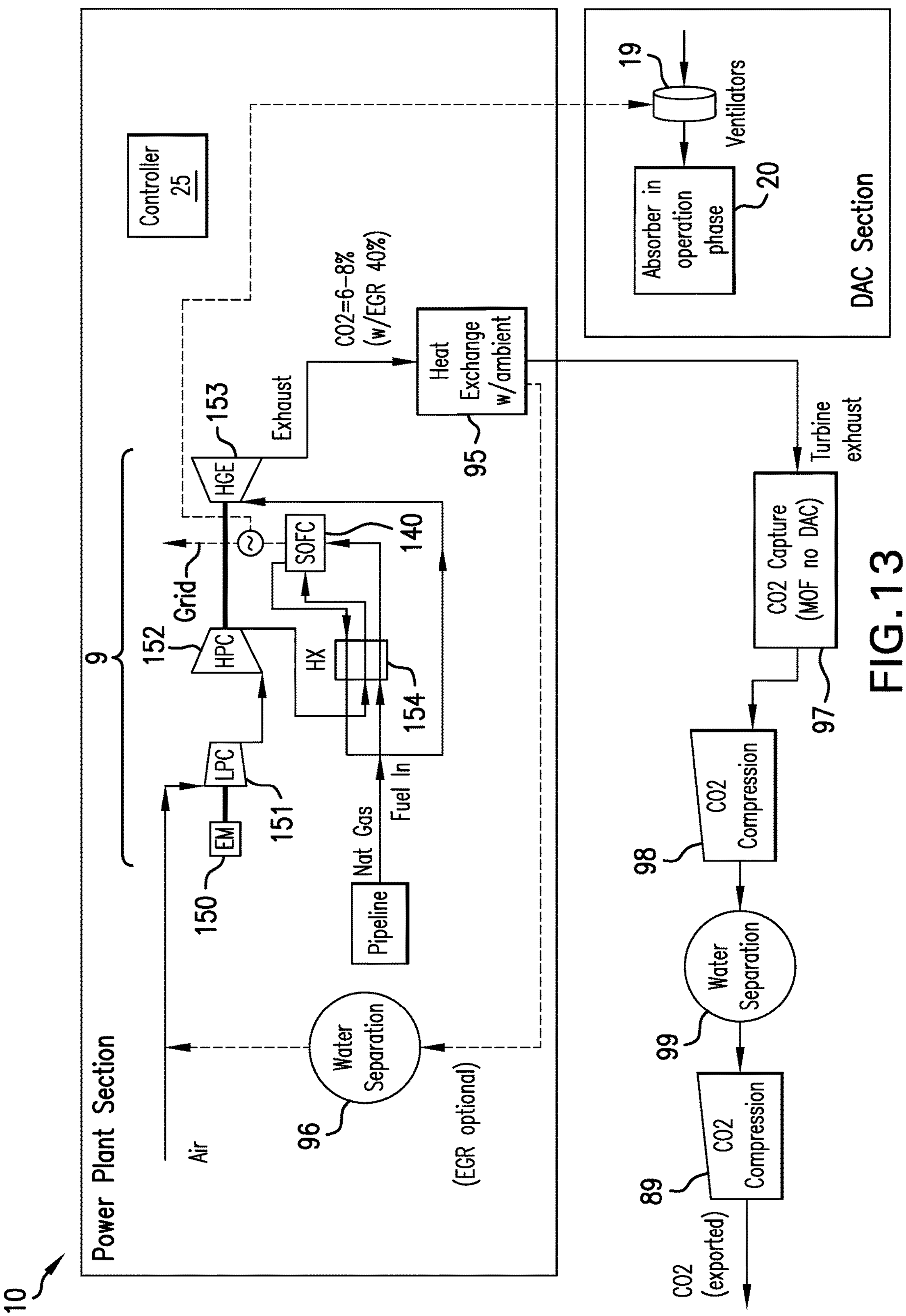

10
Power Plant Section
Controller
25
9
Air
150
EM
151
LPC
152
HPC
Grid
HGE
153
Exhaust
SOFC
140
HX
154
Pipeline
Nat Gas
Fuel In
Water
Separation
96
(EGR optional)
CO2=6-8%
(w/EGR 40%)
Heat
Exchange
w/ambient
95
DAC Section
19
Ventilators
Absorber in
operation
phase
20
Turbine
exhaust
CO2 Capture
(MOF no DAC)
97
CO2
Compression
98
Water
Separation
99
CO2
Compression
89
CO2
(exported)

FIG. 13

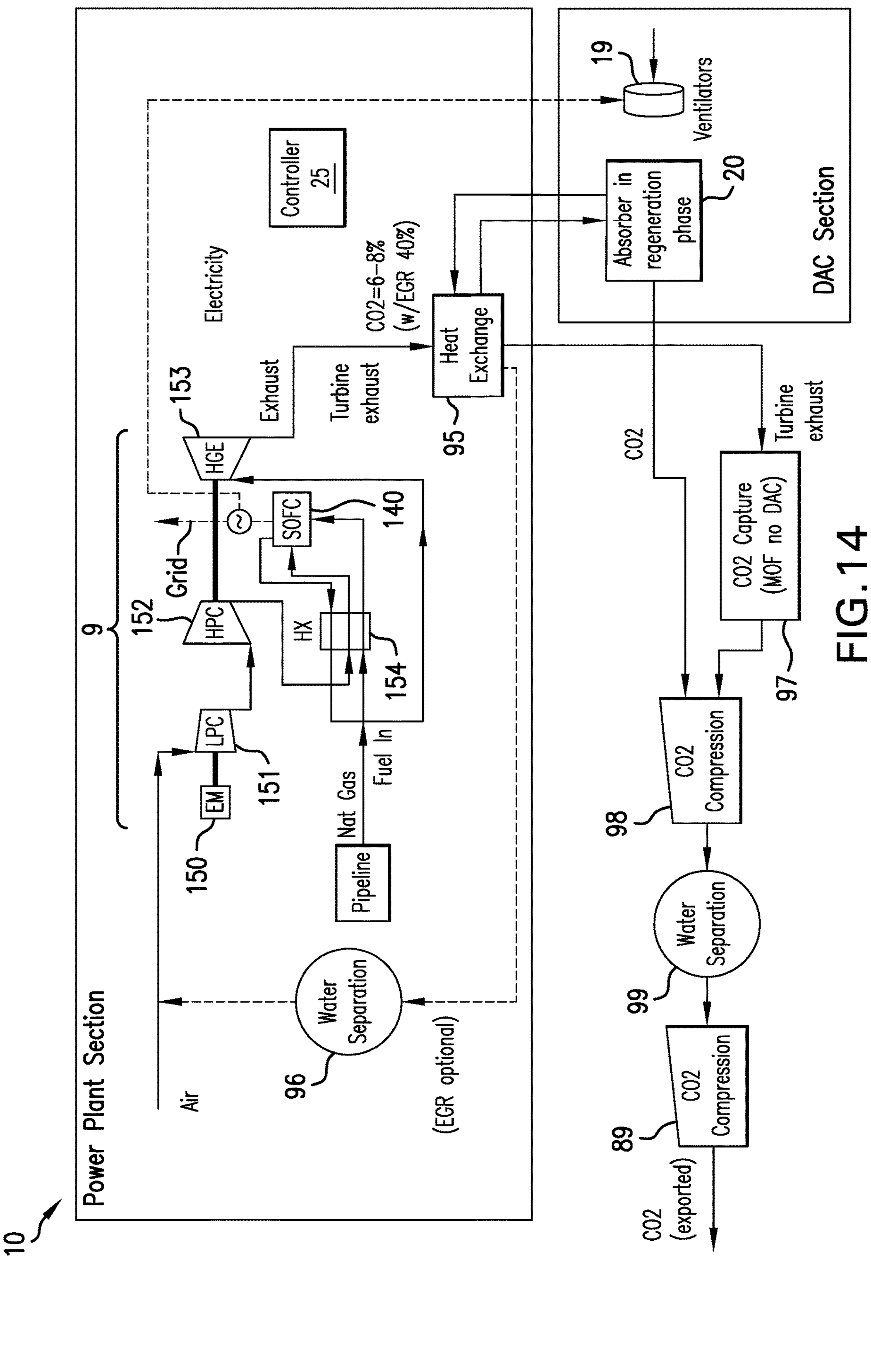

10
Power Plant Section
9
Air
EM
150
LPC
151
HPC
152
Grid
HGE
153
SOFC
140
HX
154
Pipeline
Nat Gas
Fuel In
Water Separation
96
(EGR optional)
Electricity
Controller
25
Exhaust
Turbine exhaust
CO2=6–8%
(w/EGR 40%)
Heat Exchange
95
Ventilators
19
Absorber in regeneration phase
20
DAC Section
CO2
Turbine exhaust
CO2 Capture
(MOF no DAC)
97
CO2 Compression
98
Water Separation
99
CO2 Compression
89
CO2
(exported)

FIG. 14

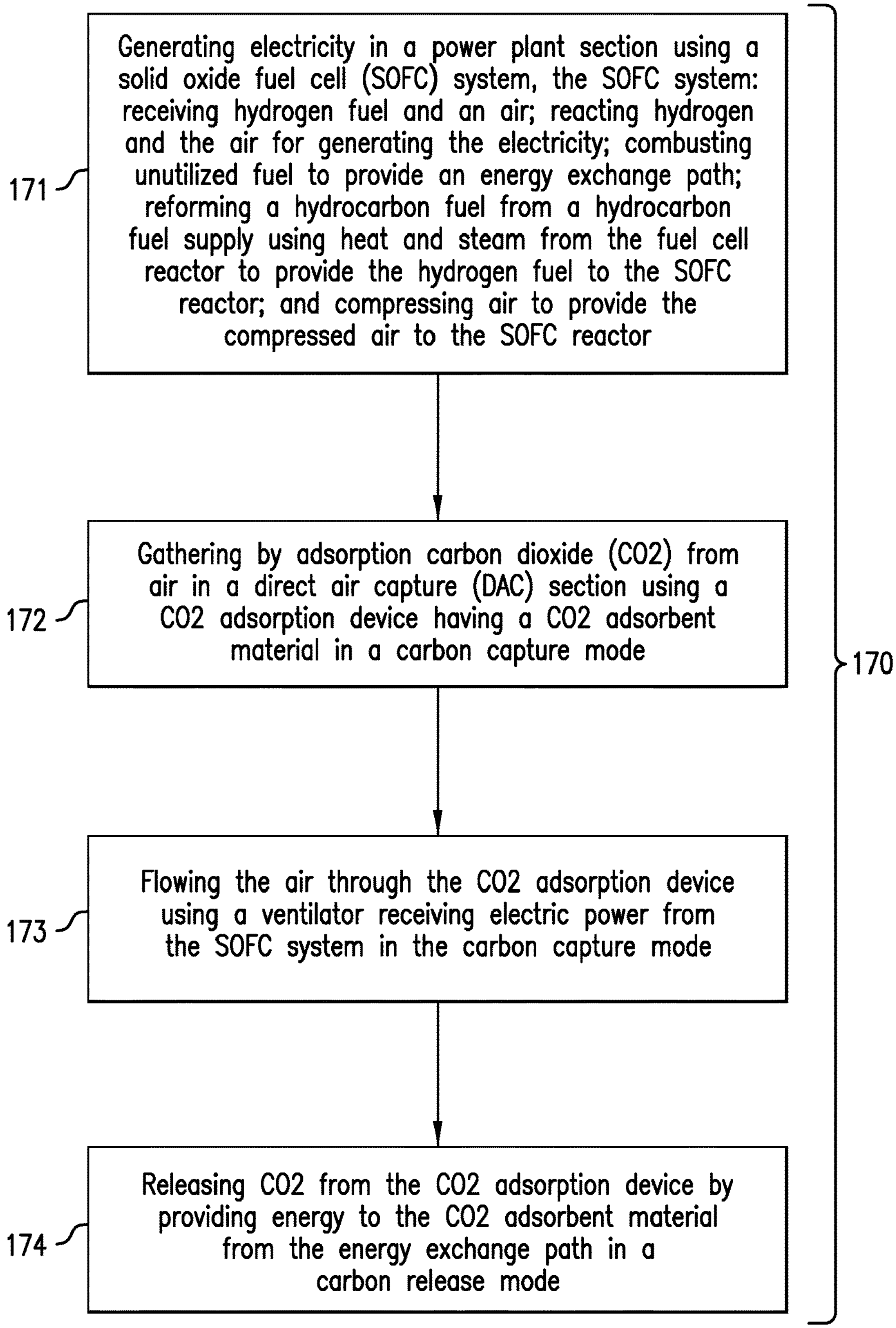

171
Generating electricity in a power plant section using a solid oxide fuel cell (SOFC) system, the SOFC system: receiving hydrogen fuel and an air; reacting hydrogen and the air for generating the electricity; combusting unutilized fuel to provide an energy exchange path; reforming a hydrocarbon fuel from a hydrocarbon fuel supply using heat and steam from the fuel cell reactor to provide the hydrogen fuel to the SOFC reactor; and compressing air to provide the compressed air to the SOFC reactor
172
Gathering by adsorption carbon dioxide (CO2) from air in a direct air capture (DAC) section using a CO2 adsorption device having a CO2 adsorbent material in a carbon capture mode
173
Flowing the air through the CO2 adsorption device using a ventilator receiving electric power from the SOFC system in the carbon capture mode
174
Releasing CO2 from the CO2 adsorption device by providing energy to the CO2 adsorbent material from the energy exchange path in a carbon release mode
170

FIG.15

DUAL PURPOSE ENERGY PLANT

BACKGROUND

This disclosure relates generally to a dual-purpose energy plant and more particularly to an energy plant that generates electricity with reduced or even negative carbon emissions as a whole.

Electricity for an electric grid generally has several power plants that combust fossil fuel to produce energy for powering an electric generator connected to the electric grid. As is generally known, the combustion process emits carbon as a byproduct of the combustion. Unfortunately, the emitted carbon can contribute to atmospheric "greenhouse" gasses that can be a contributor to so called climate change. Climate change can be potentially dangerous as causing natural disasters or economic problems, which can result from a rise in sea level or agricultural problems for example. Hence, the power industry would welcome improvements in technology to generate electricity with reduced carbon emissions compared to conventional power plants or even negative carbon emissions where the electricity generation process as a whole removes more carbon from the atmosphere than any that may be added by combustion.

BRIEF SUMMARY

Disclosed is a system for generating electricity with reduced or negative carbon emissions. The system includes: a power plant section having an electricity generating unit having an input coupled to a hydrocarbon fuel supply and an energy exchange path; and a direct air capture (DAC) section having a CO_2 adsorption device having a CO_2 adsorbent material and a ventilator electrically coupled to the electricity generating unit, the ventilator directing air flow through the CO_2 adsorption device in a carbon capture mode; wherein the CO_2 adsorption device is coupled to and in energy communication with the energy exchange path for releasing adsorbed CO_2 in a carbon release mode.

Also disclosed is a method for generating electricity with reduced or negative carbon emissions. The method includes: generating the electricity in a power plant section of an electric power plant system using an electricity generating unit being coupled to a hydrocarbon fuel supply and having an electrical output and an energy exchange path; gathering by adsorption carbon dioxide (CO_2) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO_2 adsorption device having a CO_2 adsorbent material in a carbon capture mode; flowing the air through the CO_2 adsorption device using a ventilator receiving electric power from the electricity generating unit in the carbon capture mode; and releasing CO_2 from the CO_2 adsorption device by providing energy to the CO_2 adsorbent material with energy from the energy exchange path in a carbon release mode.

BRIEF DESCRIPTION OF THE DRAWINGS

The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

FIG. **1** depicts aspects of a power plant section coupled to a direct air capture (DAC) section in a carbon capture mode;

FIG. **2** depicts aspect of a carbon dioxide adsorption device in the DAC section;

FIG. **3** depicts aspects of the power plant section coupled to the DAC section in a carbon release mode;

FIG. **4** depicts aspects of the power plant section coupled to the DAC section in the carbon release mode with the power plant section having an energy exchanger;

FIG. **5** depicts aspects of the power plant section having a supercritical CO_2 power cycle coupled to the DAC section in the carbon capture mode;

FIG. **6** depicts aspects of the power plant section having the supercritical CO_2 power cycle coupled to the DAC section in the carbon release mode;

FIG. **7** depicts aspects of the power plant section having the gas turbine and the heat recovery components coupled to the DAC section in the carbon capture mode;

FIG. **8** depicts aspects of the power plant section having the gas turbine and the heat recovery components coupled to the DAC section in the carbon release mode;

FIG. **9** depicts aspects of the power plant section having the gas turbine without the heat recovery components coupled to the DAC section in the carbon capture mode;

FIG. **10** depicts aspects of the power plant section having the gas turbine without the heat recovery components coupled to the DAC section in the carbon release mode;

FIG. **11** is a flowchart representation of a method for generating electricity with reduced or negative carbon emissions; and

FIG. **12** depicts aspect of a solid-oxide-fuel-cell (SOFC) system.

FIG. **13** depicts aspects of the SOFC system with enhancements in the carbon capture mode;

FIG. **14** depicts aspects of the SOFC system with enhancements in the carbon release mode; and

FIG. **15** is a flowchart representation of a method for generating electricity with reduced or negative carbon emissions using a fuel cell system.

DETAILED DESCRIPTION

A detailed description of one or more embodiments of the disclosed apparatus and method presented herein by way of exemplification and not limitation with reference to the figures, in which like elements are numbered alike.

Figures are presented below that illustrate various embodiments for generating electricity with reduced or negative carbon emissions. These figures include arrows for illustrating direction of fluid flow from a first component to a second component that is coupled to the first component. These arrows represent conduits, pipes, tubing, ducts or other types of flow paths for containing and directing the fluid flow. These arrows may also represent valves or dampers in the flow paths for controlling fluid flow in the flow paths in accordance with a mode of operation. These valves may be remotely controlled by a controller that may provide for automatic or manual operation. These arrows may also represent any pumps required for motivating fluid flow in accordance with a design configuration of the disclosed components. Arrows identifying electrical communication represent electrical conductors, transformers, switchgear, or other components needed for electrically powering a device. Locations where arrows leave or enter a component can represent output ports or input ports, respectively, for fluid flow or connections for electrical components. As can be seen in the figures, the arrows also indicate how a component is coupled, either directly or indirectly (with an intermediate component), to another component.

Disclosed herein are embodiments of an electric power plant system that generates electric power with reduced or negative carbon emissions. Hence, the electric power plant system has a dual purpose-generating electric power and reducing carbon emissions into the ambient air or even taking carbon out of the ambient air in total.

FIG. **1** illustrates a simplified diagram of an electric power plant system **10** having a power plant section **11** coupled to a direct air capture (DAC) section **12** operating in a carbon capture mode. The power plant section **12** includes an electricity generating unit **9** for converting a fossil fuel or hydrocarbon fuel into electricity or electrical power. Non-limiting embodiments of the electricity generating unit **9** include a supercritical carbon dioxide power cycle, a gas turbine, a reciprocating engine (e.g., a gasoline or diesel engine), or a fuel cell system that produces a carbon dioxide stream and an energy discharge stream (e.g., a solid oxide fuel cell system). Each embodiment of the electricity generating unit **9** includes an energy exchange path (e.g., a heat discharge path) that may also be referred to as an engine exhaust in some embodiments. In one or more embodiments, the electricity generating unit **9** includes a prime mover or engine **13** that combusts the hydrocarbon fuel to provide mechanical energy to drive an electric generator **14** mechanically connected (such as by the illustrated shaft) or hydraulically connected to the engine **13**. The electric generator **14** generates electricity and a first portion of the generated electricity is provided to an electric grid by a grid connection. In embodiments using the fuel cell system, an inverter (not shown) connected to electrodes of the fuel cell system may be used to supply a first portion of alternating current (AC) electricity from the fuel cell system to the grid. A second portion of the generated electricity is provided to the DAC section **12** for operation of DAC section components. A hot gas recirculation line **15** receives hot gas from the engine **13** and provides the hot gas to an energy exchanger **16**. The energy exchanger **16** provides heat energy from the hot gas to the engine **13** in a first option or provides heat energy to an ambient environment in a second option. The cooler gas from the energy exchanger **16** passes through a first water absorber or water separator **17** to remove water from the gas. Optionally, a first compressor **18** compresses the drier gas from the water separator **17** and provides the compressed gas back to the engine **13**.

The DAC section **12** includes a ventilator **19** that provides or directs ambient air to a carbon dioxide (CO_2) adsorption device **20**. The CO_2 adsorption device **20** removes CO_2 and thus the associated carbon from the ambient air. Air having less CO_2 is then discharged from the CO_2 adsorption device **20**. The ventilator **19** is electrically coupled to and receives electric power from the electricity generating unit **9**. Hereinafter, any discussion of the electricity being supplied by the electric generator **14** inherently includes electricity being supplied by the fuel cell system when used as the electricity generating unit **9**.

A controller **25** may be disposed in the power plant section **11** and/or the DAC section **12** for controlling operations of the electric power plant system **10** related to operating apparatus disclosed herein to enable electric power production with reduced or negative carbon emissions in accordance with the different modes of operation discussed herein. For example, the controller **25** may be configured to control valves or dampers for controlling a flow of fluid such as a working fluid, a heat transfer fluid, CO_2 for sequestration or export or for recycling, or air. Also, the controller **25** may be configured to control electrical switchgear for controlling power to the ventilator **19** and/or CO_2 compressors. The controller **25** may be configured to accept manual inputs and/or to provide automatic control. Automatic control may be implemented by an analog or digital processor implementing an algorithm. The algorithm may include model-based learning, machine learning, and/or artificial intelligence. In one or more embodiments, the algorithm may implement a neural network. The controller **25** may also include traditional control systems such as proportional, integral, and/or derivative (PID) control. Additionally, the controller **25** may receive inputs from sensors such as temperature, pressure, and/or flow sensors distributed throughout the electric power plant system **10**. Further, the controller **25** may be configured to communicate with other processing devices whether locally such as by wireless communication or remotely such as by over the internet. Accordingly, the controller **25** can be used to optimize the reduction of carbon emissions.

FIG. **2** depicts aspects of the CO_2 adsorption device **20**. In the embodiment of FIG. **2**, the CO_2 adsorption device **20** includes a housing **22** that supports a CO_2 adsorbent material **21** and directs air flow from the ventilator **19** to the adsorbent material **21**. Heating the CO_2 adsorbent material **21** in a carbon release mode causes the CO_2 adsorbent material **21** to release CO_2 for further processing and sequestration. In a non-limiting embodiment, the CO_2 adsorbent material **21** is a metal-organic-framework (MOF) as known in the art.

FIG. **3** illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon release mode. In the carbon release mode, heat from the energy exchanger **16** is used to heat the CO_2 adsorbent material **21** to release the adsorbed CO_2. The heat can be provided indirectly with the energy exchanger **16** being a heat exchanger or directly by mass flow of the hot exhaust or emission emitted by the electricity generating unit **9** being directed directly to the CO_2 adsorption device **20**. With the energy exchanger **16** being a heat exchanger, heat from a hot exhaust fluid on the primary side heats a heat transfer fluid on the secondary side, which is then used to heat the CO_2 adsorbent material **21**. With the energy exchanger **16** providing heat directly to the CO_2 adsorbent material **21**, the energy exchanger **16** includes components such as piping and valves to directly flow the hot exhaust fluid to the CO_2 adsorption device **20** in a mode referred to as "mass flow."

CO_2 and water discharged from the CO_2 adsorption device **20** goes to a second water absorber or water separator **30** where water is separated from the entering CO_2 and water. Dry CO_2 from the second water separator **30** goes to the first compressor **18**. A first portion of compressed CO_2 from the first compressor **18** is recycled back to the engine **13** while a second portion of the compressed CO_2 is exported for sequestration. Optionally, a carbon export heat exchanger (HX) **31** may be coupled to an output of the first compressor **18** (e.g., disposed in a pipeline conveying the second portion of the compressed CO_2 for export) to extract energy from the second portion of the compressed CO_2 for heating the CO_2 adsorbent material **21** in the carbon release mode. The second portion of the compressed CO_2 flows through a primary side of the carbon export heat exchanger **31** to heat a heat transfer fluid in the secondary side of the carbon export heat exchanger **31**. This heat transfer fluid conveys energy to the CO_2 adsorbent material **21** in the carbon release mode to aid in releasing the adsorbed CO_2.

FIG. **4** illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon release mode (also referred to as regeneration). Here, heat is provided to the CO_2 adsorber **20** to release adsorbed CO_2 by a heat exchanger **40** disposed in the DAC section **12**. A primary side of the heat exchanger **40** is coupled to and receives heat energy from the energy exchanger **16**. A secondary side of the heat exchanger **40** is coupled to and provides heat to the CO_2 adsorption device **20**. CO_2 from the second water absorber or water separator **30** is compressed for export and sequestration by a second compressor **41** disposed in the DAC section **12**. The second compressor **41** is electrically coupled to and powered by the electric generator **14**.

FIG. **5** illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon capture mode with the engine **13** being a supercritical CO_2 power cycle. In the embodiment of FIG. **5**, the supercritical CO_2 power cycle includes a combustor **50** configured to combust a hydrocarbon fuel to produce CO_2 and water in a supercritical state. In one or more embodiments, the combustor **50** includes a combustion chamber (not shown) where fuel and oxidant are combined and the combustion takes place. The supercritical CO_2 and water are provided to a turbo-expander **51**. A discharge of the turbo-expander **51** is coupled to the energy exchanger **16**, which provides a heat sink causing the supercritical CO_2 and water to expand in the turbo-expander **51** to turn an output shaft that is either directly or indirectly coupled to the electric generator **14**. In general, the turbo-expander **51** includes a turbine having blades (not shown) upon which expanding fluid impinges to turn the turbine and a turbine shaft coupled to the electric generator **14**. An air separation unit (ASU) **52** is disposed in the power plant section **11** with an output coupled to the combustor **50**. The ASU **52** is configured to separate oxygen (O_2) from ambient air to provide the separated oxygen to the combustor **50** to aid in the combustion process.

FIG. **6** illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon release mode with the engine **13** being the supercritical CO_2 power cycle. In the carbon release mode, heat is provided to the CO_2 adsorption device **20** by a heat transfer fluid in a line or pipe to the energy exchanger **16**. The heat can be provided directly by mass flow or indirectly by the energy exchanger **16** being a heat exchanger where the secondary side provides the heat. In the embodiment of FIG. **6**, the second compressor **41** provides compressed CO_2 to either an output line of the energy exchanger **16** or a separate input to the water separator **17**.

FIG. **7** illustrates a more detailed embodiment of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in the carbon capture mode with the engine **13** being a gas turbine **90** (may also be a reciprocating engine or a fuel cell inherent with electricity generating electrodes or any engine having a heat and CO_2 exhaust stream). Components not shown in either the power plant section **11** or the DAC section **12** may be considered to be disposed adjacent to the power plant section **11** or alternatively in the power plant section **11**. In the embodiment of FIG. **9**, a heat recovery power cycle (HRPC) is disposed in an exhaust path of the gas turbine **90** between an exhaust heat exchanger **95** and the gas turbine **90**. The HRPC uses waste energy from the gas turbine exhaust to generate more electricity, thus, increasing efficiency of the electric power generation. The heat recovery power cycle includes a heat recovery steam generator (HRSG) **91**, a steam turbine **92**, a HRPC heat exchanger **93**, and a HRPC pump **94**. The HRSG **91** generates steam using heat from the exhaust path of the gas turbine **90**. The generated steam is received by the steam turbine **92** to convert the energy of the steam to mechanical energy to turn a shaft coupled to the electric generator **14** or alternatively to another electric generator also connected to the electric grid. The HRPC heat exchanger **93** provides a heat sink for the discharge of the steam turbine **92** by condensing the steam to provide the necessary pressure drop across the steam turbine **92**. The HRPC pump **94** pumps the condensate from the HRPC heat exchanger **93** back to the HRSG **91** to complete the heat recovery power cycle.

Downstream of the HRPC, the exhaust heat exchanger **95** further cools the exhaust from the gas turbine **90**. The exhaust is cooled by air or other cooling source in the secondary side of the exhaust heat exchanger **95**. From the exhaust heat exchanger **95**, a first portion of the exhaust flows to a CO_2 capture device **97**, which captures CO_2 from the exhaust. In one or more embodiments, the CO_2 capture device **97** implements a Chilled Ammonia Process (CAP) or a Compact Carbon Capture with rotating bed (**3**C), all known in the art. The captured CO_2 is then compressed by a first captured CO_2 compressor **98** and dried by a captured CO_2 water separator **99**. The dried captured CO_2 is then further compressed by a second captured CO_2 compressor **89** and provided for export and sequestration. A second portion of the exhaust discharged by the exhaust heat exchanger **95** is used for exhaust gas recirculation (EGR) and flows to a water separator **96** to separate water from the second portion of the exhaust to provide dry EGR exhaust. The dry EGR exhaust is then combined with air entering the gas turbine **90** for combustion.

FIG. **8** illustrates a more detailed embodiment of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in the carbon release mode with the engine **13** being the gas turbine **90**. In the carbon release mode, both the exhaust heat exchanger **95** and the HRPC heat exchanger **93** supply heat to the CO_2 adsorption device **20** to release (also referred to as regeneration) CO_2 and supply the released CO_2 to the first captured CO_2 compressor **98**. Hence, in the carbon release mode the first captured CO_2 compressor **98** compresses CO_2 provided by the CO_2 capture device **97** and the CO_2 adsorption device **20**. In general, in the carbon release mode the controller **25** will turn off the ventilator **19** and open flow control devices to supply energy (e.g., heat) to the CO_2 adsorption device **20**.

FIG. **9** illustrates another more detailed embodiment of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in the carbon capture mode with the engine **13** being the gas turbine **90**. The embodiment of FIG. **9** is similar to the embodiment of FIG. **7**, but without the heat recovery power cycle disposed downstream of gas turbine exhaust. Here, the gas turbine exhaust goes directly to the exhaust heat exchanger **95** where the exhaust is mainly cooled before going to the CO_2 capture device **97**, the first captured CO_2 compressor **98**, the captured CO_2 water separator **99**, and the second captured CO_2 compressor **89** in that order.

FIG. **10** illustrates another more detailed embodiment of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in the carbon release mode with the engine **13** being the gas turbine **90**. Here, in the carbon release mode heat for releasing CO_2 from the adsorption device **20** is provided only or mostly by a heat transfer fluid flowing through a secondary side of the exhaust heat exchanger **95**. The heat transfer fluid is heated by heat energy in the gas turbine exhaust.

FIG. **11** is a flow chart for a method **130** for generating electricity with reduced or negative carbon emissions. Block **131** calls for generating the electricity in a power plant section of an electric power plant system using an electricity generating unit being coupled to a hydrocarbon fuel supply and having an electrical output and an energy exchange path. In one or more embodiments, the energy exchange path operates as an energy discharge path. In non-limiting embodiments, the electricity generating unit operates in a supercritical CO_2 power cycle, as a gas turbine, as a reciprocating engine, or as a fuel cell system having an energy discharge stream and a CO_2 discharge stream.

Block **132** calls for gathering by adsorption carbon dioxide (CO_2) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO_2 adsorption device having a CO_2 adsorbent material in a carbon capture mode.

Block **133** calls for flowing the air through the CO_2 adsorption device using a ventilator receiving electric power from the electricity generating unit in the carbon capture mode.

Block **134** calls for releasing CO_2 from the CO_2 adsorption device by providing energy to the CO_2 adsorbent material with energy from the energy exchange path in a carbon release mode.

In embodiments where the electricity generating unit operates in a supercritical CO_2 power cycle, the method **130** may also include: (1) combusting a hydrocarbon fuel using a combustor disposed in the supercritical CO_2 power cycle; (2) converting energy released by the combusting to mechanical output energy using an expander disposed in the supercritical CO_2 power cycle and coupled to an output of the combustor; (3) providing oxidant to the combustor using an air separation unit (ASU) coupled to an input the combustor; and (4) extracting water from working fluid flow after expansion in the expander using a water separation unit coupled to an output of the expander; wherein the electric generator is coupled to a mechanical output of the expander; and wherein the energy exchange path is a working fluid discharge path of the expander.

The method **130** may also include using an energy exchanger coupled to a working fluid discharge path of the expander for supplying energy indirectly in a form of heat or directly in a form of CO_2 mass flow to the CO_2 adsorption device in the carbon release mode to release CO_2 from the CO_2 adsorbent material, wherein the CO_2 adsorbent material comprises metal-organic-framework (MOF).

The method **130** may also include introducing a CO_2 flow from the DAC section in the carbon release mode into the combustor by using a first CO_2 compressor disposed in the DAC section and receiving electric power from the electrical generator for compressing the CO_2 released from the CO_2 adsorption material in the carbon release mode.

The method **130** may also include compressing CO_2 discharged from the energy exchanger and from the output of the first CO_2 compressor using a second CO_2 compressor disposed in the power plant section, receiving electric power from the electric generator, and coupled to an output of the water separator and an output of the first CO_2 compressor, wherein a first portion of a discharge from the second CO_2 compressor is exported for the sequestration and a second portion of the discharge from the second CO_2 compressor is recycled into the combustor.

In embodiments where the electricity generating unit includes a gas turbine, reciprocating engine, or fuel cell system each having an energy exhaust, the method **130** may include: (1) capturing CO_2 from the energy exhaust using a CO_2 capture unit coupled to the energy exhaust path; and (2) compressing the CO_2 captured by the CO_2 capture unit for exporting the compressed CO_2 using a CO_2 compressor coupled to an outlet of the CO_2 capture unit, the CO_2 compressor receiving electric power from the electricity generating unit. The CO_2 capture unit may implement a Chilled Ammonia Process (CAP) or a Compact Carbon Capture with rotating bed (**3**C). The CO_2 adsorbent material may include a metal-organic-framework (MOF). The method **130** may also include: (3) recovering heat from the energy exhaust path for generating steam using a heat recovery steam generator (HRSG) disposed in the power plant section in the energy exhaust path between the engine and the CO_2 capture unit; and (4) generating the electricity using a steam turbine coupled to an electric generator, the steam turbine receiving steam from the HRSG. The method **130** may further include: (5) compressing the CO_2 captured by the CO_2 capture unit using a first CO_2 compressor coupled to an outlet of the CO_2 capture unit, the first CO_2 compressor receiving electric power from the electricity generating unit; extracting water from compressed CO_2 from the first CO_2 compressor using a water separator coupled to an output of the first CO_2 compressor to provide dry compressed CO_2; and (6) compressing the dry compressed CO_2 using a second CO_2 compressor coupled to an output of the water separator, the second CO_2 compressor receiving electric power from the electricity generating unit.

As disclosed herein, the term "fuel cell" relates to a type of fuel cell or fuel cell system that produces a carbon dioxide stream and an energy discharge stream. One example of this type of fuel cell is a solid-oxide-fuel-cell (SOFC) system referred to as a SOFC **140** illustrated in FIG. **12**. The SOFC **140** includes a fuel cell reactor **141** that produces an electrical output to supply electricity to the grid and the DAC section **12**. Fluid outputs of the fuel cell reactor **141** are coupled to a combustor **142** and a reformer **143**. Chemical reactions for the fuel cell reactor **141** and the reformer **143** are also illustrated in FIG. **12**. The combustor **142** combusts unutilized hydrocarbon fuel from the fuel cell reactor **141**. Output from the combustor **142** provides the energy discharge stream or energy exchange path. The reformer **143** performs a steam reforming reaction using heat and steam from the fuel cell reactor **141** to reform a hydrocarbon fuel to provide hydrogen to the fuel cell reactor **141** where the hydrogen is oxidized at an electrode to produce electricity.

FIG. **13** illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon capture mode with the electricity generating unit **9** being the SOFC **140** integrated with enhancements that include an expander. In this embodiment, the electricity generating unit **9** includes the SOFC system **140**, an electric motor (EM) **150**, a low pressure (LP) air compressor **151**, a high pressure (HP) compressor **152**, an expander **153**, and a multi-stream heat exchanger (HX) **154**. The EM **150** is coupled to the LP compressor **151** for driving the LP compressor **151** and receives electric power from the SOFC system **140**. During start-up a local battery may be used to temporarily power the EM **150** until the fuel cell reactor begins to generate the electricity. Compressed air from the LP compressor **151** is provided to an input of the HP compressor **152**. The terms LP and HP may be considered as relative terms such that the pressure output of the HP compressor **152** is higher than the pressure output of the LP compressor **151**. The expander **153** is coupled to the HP compressor **152** for driving the HP compressor **152**. The multi-stream heat exchanger (HX) **154** has a primary side and at least two secondary sides heated by the primary side. A primary side input is coupled to a high pressure and high temperature exhaust stream of the SOFC system **140**, while a primary side output is coupled to an input of the expander **153**. Hence, the high pressure and high temperature exhaust stream of the SOFC system **140** powers the expander **153** and thus the HP compressor **152**. One secondary side of the HX **154** is coupled to a hydrocarbon fuel input to the SOFC system **140** for heating the hydrocarbon fuel to the SOFC **140**. Another secondary side of the HX **154** is coupled to a HP output of the HP compressor **152** for heating an oxidant supply to the SOFC system **140**.

The SOFC system **140** integrated with the above components are configured to produce electric power that is provided to the grid, to the EM **150**, and to the DAC components. The SOFC system **140** produces electrical power when a fuel and oxidant undergo an electrochemical reaction inside the SOFC **140** under certain defined conditions. The expander produces mechanical power by converting work of high pressure and temperature exhaust gas of the SOFC system **140** that is expanded to low temperature and low pressure. The LP and HP compressors **151** and **152** are configured to deliver high pressure oxidant (e.g., air) to the SOFC system **140**. The SOFC system **140** is configured to receive compatible hydrocarbon fuels such as natural gas for example. The heat exchanger **154** is external to the SOFC system **140** and is configured to allow heat exchange between 3 streams—(1) a fuel stream of a SOFC fuel feed line, upstream the SOFC **140**, (2) the exhaust gas from the SOFC **140**, downstream of SOFC **140**, and (3) a high-pressure oxidant feed line to the SOFC **140**, upstream of SOFC **140**.

The electrical power is generated by the pressurized fuel cell reactor **141** performing an electrochemical reaction between pre-heated fuel and oxidant streams coming from the multi-stream heat exchanger **154**. A combustion chamber of the combustor **142** is an integral part of SOFC **140**, that helps in burning unutilized fuel from the pressurized fuel cell reactor **141** by use of oxidant, producing high temperature exhaust gases. The hot exhaust gases from the fuel cell reactor **141** exchange heat with fuel and oxidant streams in the multi-stream heat exchanger **154**. The expander **153** is arranged down stream of heat exchanger **154** and is driven by hot exhaust gases of SOFC system **140**. The expander **153** and HP compressor **152** are connected to each other through a common shaft in one or more embodiments. The power produced by the expander **153** is used to drive the HP compressor **152** and excess power may be extracted through a generator (not shown) connected to the other side of the shaft of the expander **153**.

The power producing unit, the solid oxide fuel cell system **140** integrated with the expander **153** operates as follows. Air to the SOFC system **140** is compressed through a two-stage compression system. An air stream is drawn from atmosphere by the low-pressure compressor **151**, which is driven by the small electrical motor **150**, is compressed to a first stage of pressure. The air from the first stage of compression is fed to the high-pressure compressor **152**, which is driven by the gas expander **153**, and the air is further compressed to high pressure. A compressed air stream from the HP compressor **152** is passed through the heat exchanger **154** before being introduced in to SOFC system **140**. The heat exchanger **154** also heats a fuel stream, the source of power, from an outside source. The heated stream of air and the heated stream of fuel from the heat exchanger are injected in to the SOFC system **140**. Inside the fuel cell reactor **141**, the heated air and the heated fuel undergo an electro-chemical reaction at the electrodes of the fuel cell reactor **141** to produce electrical power. The exhaust gases, unutilized fuel and air are mixed in the integral combustor **142** of the SOFC **140** and combusted in the combustion chamber. This energy of combustion further increases the temperature of the exhaust gas stream from the combustor **142**. The high temperature exhaust gas stream is then fed to the heat exchanger **154** to exchange heat with incoming streams of fuel and air. From the heat exchanger **154**, the high pressure and high temperature exhaust gas stream is fed to the expander **153**. The high pressure and high temperature exhaust gas stream undergoes expansion in the expander **153** converting its energy to work and further optionally to electrical power. The expander **153** transmits part power to HP compressor **152** and optionally remaining power to a generator through connected common shaft.

FIG. **14** illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon release mode with the electricity generating unit **9** being the SOFC system **140** integrated with the expander **153**. In the carbon release mode, energy (i.e., heat) is directed from the heat exchanger **95** to the CO_2 adsorption device **20** to heat the CO_2 adsorption device **20**. The CO_2 released from the CO_2 adsorption device **20** is directed to the first captured CO_2 compressor **98** for further processing prior to being exported for sequestration.

FIG. **15** is a flow chart for a method **170** for generating electricity with reduced or negative carbon emissions using a fuel cell system. Block **171** calls for generating the electricity in a power plant section using a solid oxide fuel cell (SOFC) system, the SOFC system: receiving hydrogen fuel and an air; reacting hydrogen and the air for generating the electricity; combusting unutilized fuel to provide an energy exchange path; reforming a hydrocarbon fuel from a hydrocarbon fuel supply using heat and steam from the fuel cell reactor to provide the hydrogen fuel to the SOFC reactor; and compressing air to provide the compressed air to the SOFC reactor. In one or more embodiments, the energy exchange path operates as an energy discharge path.

Block **172** calls for gathering by adsorption carbon dioxide (CO_2) from air in a direct air capture (DAC) section using a CO_2 adsorption device having a CO_2 adsorbent material in a carbon capture mode.

Block **173** calls for flowing the air through the CO_2 adsorption device using a ventilator receiving electric power from the SOFC system in the carbon capture mode.

Block **174** calls for releasing CO_2 from the CO_2 adsorption device by providing energy to the CO_2 adsorbent material from the energy exchange path in a carbon release mode. The releasing may include at least one of heating the CO_2 adsorbent material using mass flow of a fluid in the energy exchange path or heating the CO_2 adsorbent material using a heat transfer fluid heated by a heat exchanger in a secondary side, a primary side of the heat exchanger being heated by energy from the energy exchange path.

The method **170** may also include capturing CO_2 using a CO_2 capture device receiving an exhaust from the combustor.

The method **170** may also include: compressing CO_2 released by the CO_2 adsorbent material and the CO_2 capture device to provide compressed CO_2; separating water from the compressed to provide dry compressed CO_2; compressing the dry compressed CO_2 to provide further compressed dry CO_2; and exporting the further compressed dry CO_2.

The method **170** may also include: driving a high-pressure compressor providing the compressed air to the SOFC reactor using an expander coupled to the high-pressure compressor and driven by exhaust from the combustor; and reforming the hydrocarbon fuel using energy from the SOFC reactor.

The method **170** may also include providing low-pressure compressed air to the high-pressure compressor using a low-pressure compressor; and driving the low-pressure compressor using an electric motor coupled to the low-pressure compressor.

The method **170** may also include: drying exhaust from the expander to provide dried exhaust; and recycling the dried exhaust into the low-pressure compressor.

In support of the teachings herein, various analysis components may be used, including a digital and/or an analog system. For example, the controller **25** may include digital and/or analog systems. The system may have components such as a processor, storage media, memory, input, output, communications link (wired, wireless, optical or other), user interfaces (e.g., a display or printer), software programs, signal processors (digital or analog) and other such components (such as resistors, capacitors, inductors and others) to provide for operation and analyses of the apparatus and methods disclosed herein in any of several manners well-appreciated in the art. It is considered that these teachings may be, but need not be, implemented in conjunction with a set of computer executable instructions stored on a non-transitory computer readable medium, including memory (ROMs, RAMs), optical (CD-ROMs), or magnetic (disks, hard drives), or any other type that when executed causes a computer to implement the method of the present invention. These instructions may provide for equipment operation, control, data collection and analysis and other functions deemed relevant by a system designer, owner, user or other such personnel, in addition to the functions described in this disclosure.

Batteries to supply electric power for start-up or shutdown purposes may be disposed in various locations throughout the electric power plant system **10**. The batteries may be charged using electricity generated in the power plant section **11**.

Set forth below are some embodiments of the foregoing disclosure:

Embodiment 1. A system for generating electricity with reduced or negative carbon emissions, including a power plant section comprising an electricity generating unit having an input coupled to a hydrocarbon fuel supply and an energy exchange path, a direct air capture (DAC) section comprising a CO_2 adsorption device having a CO_2 adsorbent material and a ventilator electrically coupled to the electricity generating unit, the ventilator directing air flow through the CO_2 adsorption device in a carbon capture mode, wherein the CO_2 adsorption device is coupled to and in energy communication with the energy exchange path for releasing adsorbed CO_2 in a carbon release mode.

Embodiment 2. The system as in any prior embodiment, further including a CO_2 compressor coupled to a CO_2 release port of the CO_2 adsorption device for compressing a portion of the adsorbed CO_2 in a carbon release mode and providing the portion for export, and a carbon export heat exchanger having a primary side coupled to an output of the CO_2 compressor and a secondary side coupled to the CO_2 adsorption device for heating the CO_2 adsorbent material in the carbon release mode.

Embodiment 3. The system as in any prior embodiment, wherein the electricity generating unit comprises components of a supercritical CO_2 power cycle, a gas turbine, a reciprocating engine, or a fuel cell system producing a CO_2 stream and an energy discharge stream.

Embodiment 4. The system as in any prior embodiment, wherein the components of the supercritical CO_2 power cycle comprise a combustor disposed in the thermodynamic power cycle to combust a hydrocarbon fuel, an air separation unit (ASU) coupled to an input the combustor for providing oxidant to the combustor, an expander coupled to an output of the combustor for converting energy released by the combusting to mechanical output energy and comprising a working fluid discharge path as the energy exchange path, an electric generator coupled to the expander to receive the mechanical output energy from the expander, and a water separation unit coupled to the working fluid discharge path of the expander to extract water from working fluid flow after expansion of the working fluid in the expander.

Embodiment 5. The system as in any prior embodiment, wherein the components further comprise an energy exchanger coupled to the working fluid discharge path of the expander to supply energy indirectly in a form of heat from a heat exchanger or directly in a form CO_2 mass flow to the CO_2 absorption device in the carbon release mode to release CO_2 from the CO_2 absorbent material.

Embodiment 6. The system as in any prior embodiment, further including a first CO_2 compressor disposed in the DAC section, electrically coupled to the electric generator, having an input in fluid communication with the CO_2 released from the CO_2 adsorbent material in the carbon release mode to compress the released CO_2 to provide compressed CO_2, and having an output coupled to the supercritical CO_2 power cycle to introduce the compressed CO_2 into the supercritical CO_2 power cycle.

Embodiment 7. The system as in any prior embodiment, further including a second CO_2 compressor disposed in the power plant section, electrically coupled to the electric generator, having an input coupled to an output of the water separator and an output of the first CO_2 compressor to compress CO_2 discharged from the energy exchanger and from the first CO_2 compressor, and having an output coupled to a carbon sequestration process and to the supercritical CO_2 power cycle to recycle a portion of the CO_2.

Embodiment 8. The system as in any prior embodiment, wherein the CO_2 adsorbent material comprises a metal-organic-framework (MOF).

Embodiment 9. The system as in any prior embodiment, wherein the electricity generating unit comprises the gas turbine, the reciprocating engine, or the fuel cell system, each having a combustion exhaust path as the energy exchange path, the system further including a CO_2 capture unit coupled to the combustion exhaust path for capturing CO_2 from the combustion exhaust path, and a CO_2 compressor and coupled to an outlet of the CO_2 capture unit for compressing the CO_2 captured by the CO_2 capture unit.

Embodiment 10. The system as in any prior embodiment wherein the CO_2 capture unit comprises a chilled ammonia process (CAP) or a compact carbon capture with rotating bed (**3**C) and the CO_2 adsorbent material comprises a metal-organic-framework (MOF).

Embodiment 11. The system as in any prior embodiment, further including components of a heat recovery power cycle, the components including a heat recovery steam generator (HRSG) disposed in the power plant section in the combustion exhaust path downstream of an exhaust port of the electricity generating unit and upstream of the CO_2 capture unit, the HRSG generating steam from heat of the engine exhaust path, a steam turbine coupled to the HRSG for receiving the generated steam and having a mechanical output coupled to the electric generator or another electric generator for generating the electricity, and an energy recovery energy exchanger disposed in the heat recovery power cycle downstream of the steam turbine and in energy communication with the CO_2 adsorption device during the carbon release mode.

Embodiment 12. The system as in any prior embodiment, wherein the fuel cell system includes a fuel cell reactor that performs an electro-chemical reaction of hydrogen and air to generate the electricity, the fuel cell reactor having an input that receives the hydrogen and air, an exhaust output that discharges unutilized hydrogen, a steam exhaust output that discharges steam, and a heat exhaust output that discharges energy, a combustor coupled to the exhaust output and the steam exhaust output, the combustor combusting the unutilized hydrogen, a combustion exhaust path of the combustor being the energy exchange path, and a reformer coupled to the heat exhaust output and the steam exhaust output of the fuel cell reactor, the reformer using the energy and the steam to reform a hydrocarbon fuel to provide the hydrogen.

Embodiment 13. A method for generating electricity with reduced or negative carbon emissions, the method including generating the electricity in a power plant section of an electric power plant system using an electricity generating unit being coupled to a hydrocarbon fuel supply and having an electrical output and an energy exchange path, gathering by adsorption carbon dioxide (CO_2) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO_2 adsorption device having a CO_2 adsorbent material in a carbon capture mode, flowing the air through the CO_2 adsorption device using a ventilator receiving electric power from the electricity generating unit in the carbon capture mode, and releasing CO_2 from the CO_2 adsorption device by providing energy to the CO_2 adsorbent material with energy from the energy exchange path in a carbon release mode.

Embodiment 14. The method as in any prior embodiment, further including compressing the CO_2 released from the CO_2 adsorption device using a CO_2 compressor to provide compressed CO_2 for export, extracting heat from the CO_2 for export using a carbon export heat exchanger coupled to the CO_2 compressor to provide extracted heat, and heating the CO_2 adsorbent material with the extracted heat to release CO_2 from the CO_2 adsorption device in the carbon release mode.

Embodiment 15. The method as in any prior embodiment, wherein the electricity generating unit operates in a supercritical CO_2 power cycle, as a gas turbine, as a reciprocating engine, or a fuel cell system producing a carbon dioxide stream and an energy discharge stream.

Embodiment 16. The method as in any prior embodiment, wherein the electricity generating unit operates in the supercritical CO_2 power cycle and the method further includes combusting a hydrocarbon fuel using a combustor disposed in the supercritical CO_2 power cycle, converting energy released by the combusting to mechanical output energy using an expander disposed in the supercritical CO_2 power cycle and having an input coupled to an output of the combustor, the expander being coupled to an electric generator for generating the electricity, providing oxidant to the combustor using an air separation unit (ASU) coupled to an input the combustor, extracting water from working fluid flow after expansion in the expander using a water separation unit coupled to an output of the expander, wherein the energy exchange path is a working fluid discharge path of the expander.

Embodiment 17. The method as in any prior embodiment, further including using an energy exchanger coupled to the working fluid discharge path of the expander for supplying energy indirectly in a form of heat or directly in a form of CO_2 mass flow to the CO_2 adsorption device in the carbon release mode to release CO_2 from the CO_2 adsorbent material, wherein the CO_2 adsorbent material comprises metal-organic-framework (MOF).

Embodiment 18. The method as in any prior embodiment, further including introducing a CO_2 flow from the DAC section in the carbon release mode into the combustor by using a first CO_2 compressor disposed in the DAC section and receiving electrical power from the electric generator for compressing the CO_2 released from the CO_2 adsorption material in the carbon release mode.

Embodiment 19. The method as in any prior embodiment, further including compressing CO_2 discharged from the energy exchanger and from the output of the first CO_2 compressor using a second CO_2 compressor disposed in the power plant section, receiving electric power from the electric generator, and coupled to an output of the water separator and an output of the first CO_2 compressor, wherein a first portion of a discharge from the second CO_2 compressor is exported for the sequestration and a second portion of the discharge from the second CO_2 compressor is recycled into the combustor.

Embodiment 20. The method as in any prior embodiment, wherein the electricity generating unit operates as the gas turbine, the reciprocating engine, or the fuel cell system, each having a combustion exhaust path as the energy exchange path, the method further including capturing CO_2 from the combustion exhaust path using a CO_2 capture unit coupled to the combustion exhaust path, and compressing the CO_2 captured by the CO_2 capture unit for exporting the compressed CO_2 using a CO_2 compressor coupled to an outlet of the CO_2 capture unit, the CO_2 compressor receiving electric power from the electric generating unit.

Embodiment 21. The method as in any prior embodiment, wherein the CO_2 capture unit comprises a Chilled Ammonia Process (CAP) or a Compact Carbon Capture with rotating bed (**3**C), and the CO_2 adsorbent material includes a metal-organic-framework (MOF).

Embodiment 22. The method as in any prior embodiment, further including recovering energy from the combustion exhaust path for generating steam using a heat recovery steam generator (HRSG) disposed in the power plant section in the combustion exhaust path between the electricity generating unit and the CO_2 capture unit, and generating the electricity using a steam turbine coupled to the electric generator or another electric generator, the steam turbine receiving steam from the HRSG.

Embodiment 23. The method as in any prior embodiment, wherein the compressing the CO_2 captured by the CO_2 capture unit includes compressing the CO_2 captured by the CO_2 capture unit using a first CO_2 compressor coupled to an outlet of the CO_2 capture unit, the first CO_2 compressor receiving electric power from the electricity generating unit, extracting water from compressed CO_2 from the first CO_2 compressor using a water separator coupled to an output of the first CO_2 compressor to provide dry compressed CO_2, and compressing the dry compressed CO_2 using a second CO_2 compressor coupled to an output of the water separator, the second compressor receiving electric power from the electricity generating unit.

Elements of the embodiments have been introduced with either the articles "a" or "an." The articles are intended to mean that there are one or more of the elements. The terms "including" and "having" and the like are intended to be inclusive such that there may be additional elements other than the elements listed. The conjunction "or" when used with a list of at least two terms is intended to mean any term or combination of terms. The term "configured" relates to one or more structural limitations of a device that are required for the device to perform the function or operation for which the device is configured. The term "coupled" relates to being coupled directly or indirectly using an intermediate device. The terms "first" and "second" and the like are used to distinguish terms and not to denote a particular order.

The flow diagram depicted herein is just an example. There may be many variations to this diagram or the steps (or operations) described therein without departing from the scope of the invention. For example, operations may be performed in another order or other operations may be performed at certain points without changing the specific disclosed sequence of operations with respect to each other. All of these variations are considered a part of the claimed invention.

The disclosure illustratively disclosed herein may be practiced in the absence of any element which is not specifically disclosed herein.

While one or more embodiments have been shown and described, modifications and substitutions may be made thereto without departing from the scope of the invention. Accordingly, it is to be understood that the present invention has been described by way of illustrations and not limitation.

It will be recognized that the various components or technologies may provide certain necessary or beneficial functionality or features. Accordingly, these functions and features as may be needed in support of the appended claims and variations thereof, are recognized as being inherently included as a part of the teachings herein and a part of the invention disclosed.

While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited.

What is claimed is:

1. A system for generating electricity provided to an electric grid with reduced or negative carbon emissions, the system comprising:

- a power plant section comprising an electricity generating unit comprising an energy exchange path, wherein the electricity generating unit comprises:
  - a working fluid comprising carbon dioxide (CO_2);
  - a combustor coupled to a supply of hydrocarbon fuel and a supply of pure oxygen and configured to combust the hydrocarbon fuel with the pure oxygen to heat the CO_2 to a supercritical state and to produce additional CO_2;
  - an expander coupled to an output of the combustor to receive the CO_2 in the supercritical state and configured to convert energy of the CO_2 in the supercritical state to mechanical output energy, the expander comprising a working fluid discharge path as the as the energy exchange path;
  - an electric generator coupled to the expander and configured to receive the mechanical output energy for generating the electricity, the electric generator comprising an output coupled to the electric grid;
  - an energy exchanger comprising an input coupled to the expander and configured to receive CO_2 discharged from the expander;
  - a compressor comprising an input coupled to an output of the energy exchanger and an output coupled to an input of the combustor to compress and recycle at least a portion of the CO_2 discharged by the expander;
- a direct air capture (DAC) section comprising a CO_2 adsorption device having a CO_2 adsorbent material and a ventilator electrically coupled to the electric generator, the ventilator directing air flow through the CO_2 adsorption device in a carbon capture mode;
- wherein the CO_2 adsorption device is coupled to and in energy communication with the energy exchange path for releasing adsorbed CO_2 in a carbon release mode further comprising a water separator disposed in a line between the energy exchanger and the compressor and configured to separate water from a discharge of the energy exchanger; further comprising another compressor disposed in a line between the CO_2 adsorption device and at least one of an input of the water separator or a line leading to an input of the water separator, the another compressor comprising an input coupled to an output of the CO_2 adsorption device configured to discharge CO_2 in the carbon release mode.

2. The system according to claim **1**, further comprising:

- a CO_2 compressor coupled to a CO_2 release port of the CO_2 adsorption device for compressing a portion of the adsorbed CO_2 in a carbon release mode and providing the portion for export; and
- a carbon export heat exchanger having a primary side coupled to an output of the CO_2 compressor and a secondary side coupled to the CO_2 adsorption device for heating the CO_2 adsorbent material in the carbon release mode.

3. The system according to claim **1**, wherein the CO_2 adsorbent material comprises a metal-organic-framework (MOF).

4. The system according to claim **1**, wherein the CO_2 adsorption device is coupled to the working fluid discharge path of the expander such that the CO_2 adsorbent material is heated in the carbon release mode by CO_2 discharged from the expander.

5. The system according to claim **1**, further comprising a heat exchanger having a first side coupled to the working fluid discharge path of the expander and a second side coupled to the CO_2 adsorption device such that the CO_2 adsorbent material is heated in the carbon release mode by a heat transfer fluid heated in the heat exchanger by CO_2 discharged from the expander.

6. A method for generating electricity provided to an electric grid with reduced or negative carbon emissions, the method comprising:

- using carbon dioxide (CO_2) as a working fluid in an electricity generating unit of a power plant section of an electric power plant system;
- combusting hydrogen fuel and pure oxygen in a combustor coupled to a supply of hydrocarbon fuel and a

supply of pure oxygen to heat the CO_2 to a supercritical state and to produce additional CO_2;

converting energy of the CO_2 in the supercritical state to mechanical output energy using an expander coupled to an output the combustor;

generating the electricity using an electric generator coupled to the expander to receive the mechanical output energy and to provide the electricity to the electric grid;

using a working fluid discharge path of the expander as an energy exchange path;

receiving CO_2 discharged from the expander using an energy exchanger comprising an input coupled to the expander to receive CO_2 discharged from the expander;

compressing and recycling at least a portion of the CO_2 discharged by the expander to the combustor using a compressor comprising an input coupled to an output of the energy exchanger and an output coupled to an input of the combustor;

gathering by adsorption carbon dioxide (CO_2) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO_2 adsorption device having a CO_2 adsorbent material in a carbon capture mode;

flowing the air through the CO_2 adsorption device using a ventilator receiving electric power from the electric generator in the carbon capture mode; and

releasing CO_2 from the CO_2 adsorption device by providing energy to the CO_2 adsorbent material with energy from the energy exchange path in a carbon release mode further comprising separating water from a discharge comprising CO_2 and water from the energy exchanger using a water separator having an output coupled to the input of the compressor; further comprising compressing CO_2 output from the CO_2 adsorption device in the carbon release mode using another compressor and providing compressed CO_2 from the another compressor to at least one of an input of the water separator or a line leading to an input of the water separator.

7. The method according to claim **6**, further comprising:

compressing the CO_2 released from the CO_2 adsorption device using a CO_2 compressor to provide compressed CO_2 for export;

extracting heat from the CO_2 for export using a carbon export heat exchanger coupled to the CO_2 compressor to provide extracted heat; and

heating the CO_2 adsorbent material with the extracted heat to release CO_2 from the CO_2 adsorption device in the carbon release mode.

8. The method according to claim **6**, wherein releasing CO_2 from the CO_2 adsorption device by providing energy to the CO_2 adsorbent material with energy from the energy exchange path in a carbon release mode comprises heating the CO_2 adsorbent material with CO_2 discharged from the expander.

9. The method according to claim **6**, wherein releasing CO_2 from the CO_2 adsorption device by providing energy to the CO_2 adsorbent material with energy from the energy exchange path in a carbon release mode comprises heating the CO_2 adsorbent material with a heat transfer fluid heated in a second side of a heat exchanger that receives CO_2 discharged from the expander in a first side of the heat exchanger.

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